

Precalculus

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Dedication

These notes are dedicated to the love of my life, Bonnie S. Howard.

To the Instructor

These notes are a self-contained sequence of problems designed for a college precalculus course using an inquiry based learning pedagogy. They were developed over four complete semesters and two summer sessions taught at a private mid-western university with an undergraduate enrollment of 2200 students. Enrollment in the four semesters was 29, 11, 28 and 29 with DFW rates 21%, 9%, 21% and 7% respectively. Enrollment in the two summer sessions was 5 and 4 with DFW rates 20% and 25% respectively. Thus, over the course of four full semesters and two summer sessions the enrollment was 106 with a DFW rate of 16%. The semesters were typical 15 weeks long with the class meeting four times per week for 50 minutes for each class period.

This work began as a direct extension of Dr. Ted Mahavier's Trigonometry notes obtained from JIBLM (2007). Chapters 3, 4 and 5 are also a direct extension of Mahavier's work as they were developed from his Calculus *I, II & III* notes, JIBLM (2015). The problems contained in this work have been well tested in classroom settings. Working with class sizes over an enrollment of 25 does take some extra care. I have found that, especially with larger class sizes, it is best to divide the work into 15 – 20 problems per week. Each weekend the students are expected to study the set number of problems from the previous week, expecting a quiz on Monday covering these problems plus any needed algebra reviews. This does require strategically placed mini-lectures to insure that all problems from the week get discussed in class.

These notes break from the traditional view for a precalculus course where algebraic topics are covered first, followed by trigonometry. These notes begin with a brief review of some graphing methods followed by jumping directly into trigonometry. There are a few reasons for this pedagogical choice. First, most students coming into the course have studied at least Algebra 1 and 2 at the high school level. To a certain degree, they are held accountable for this expected prior knowledge. Secondly, when deficiencies

arise, the students are forced to realize the significance of these shortcomings. The goal is to force students to understand the struggle they will run into in Calculus if they do not address their algebraic weaknesses. This does require instructors to keep a very close eye on what algebraic weaknesses need to be addressed. A few key areas I have noticed consistently: 1). Wheel motion problems reveal proportionality weaknesses as well as the ability to coordinate units. 2). Trigonometric identities reveal weaknesses working with complex fractions. In any case, it is very important for instructors to notice when there are algebraic weaknesses and to address them accordingly.

When algebraic weaknesses are revealed, they should be dealt with in the context where students understand that their weaknesses are hindering future mathematical knowledge. Mini-lectures, class discussions and quizzes, give direct help in overcoming algebraic weaknesses. These weaknesses are always distinct for each class. Most importantly, one-on-one discussions lets students know that they need to take serious responsibility to work on their algebraic weaknesses, including seeking outside help. Otherwise they will struggle in a Calculus series.

By dealing with algebraic weaknesses within the context of studying trigonometry, I have found that covering trigonometry topics does take a little longer. Generally, the normal algebraic topics (exponents, logarithms, polynomials) come along a little quicker. Using these notes without keenly identifying algebraic weaknesses would be problematic. I would be more than happy to share quizzes and other review materials as well as advice on how these notes have revealed algebraic weaknesses in my own experiences.

Throughout the notes students are asked to prove various statements or theorems. Student's first efforts usually fall short of what would be considered a rigorous proof. After any student attempts to write a proof it is quite important for the instructor to point out areas of their proof that lack rigor and to explicitly modify the students work to illustrate what rigorous proofs should look like. This will help the class learn how to write rigorous proofs. As the class progresses the hope is that student proofs become more rigorous.

Footnotes to the instructor have been referenced throughout the text and can be found at the end of the notes.

Please feel free to send me an email with questions and/or concerns.

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To the Student

Welcome to Pre-calculus! The structure of this course will likely be quite different than any previous mathematics course you have taken. There will be no traditional textbook. The notes that you develop as you and your classmates work through a series of problems will be your book. Problems will be worked out and presented in class by you and your classmates. You will be actively involved in doing mathematics rather than being told how to do mathematics. The problems in these notes were first developed by Dr. Ted Mahavier of Lamar University and have been modified for this course.

The process of doing mathematics involves thinking through ideas and trying to find solutions to problems. This is usually a very messy process. When a textbook author or an instructor presents ideas, concepts and problems to students, they have typically spent lots of time polishing their thoughts. Hidden to the student is the messy work of seeking to understand the ideas and problems in order to bring a nice concise presentation to the students. In this class you will not be protected from the messy process of doing mathematics. Nor will you be protected from the process of presenting clear and precise solutions to mathematical problems.

To accomplish the goals of this course it will be very important for you as a student to keep records of both the messy processes of you doing mathematics and your precise articulation of neatly organized mathematics. Thus, in addition to presentations, quizzes and exams there will be two major requirements for this course:

1. Each student must keep one notebook separate from all other courses they may be taking. This notebook will be a record of the messy process of doing mathematics. This notebook must contain all class notes including notes of all presentations. This notebook must also contain a written record of all attempts to the problems and theorems worked on throughout the course. The notebook will be collected and graded

twice during the semester.

VERY IMPORTANT NOTES REGARDING ORGANIZATION OF NOTEBOOK:

As noted above, your notebook must be used exclusively for this class. It is expected that this notebook will contain messy mathematics. The notebook must contain all class notes and all attempts of problems and theorems. All entries must be dated using the following format: A new page must be started for each day entries are made. Write and underline the date at the top of the new page. This is to be followed by all entries for the day, including class notes as well as problem attempts. There is no need to date every page, just the starting page for each day you work in the notebook. Entries for another day must start a new page. **VERY IMPORTANT:** If you miss class for any reason, you must copy notes from a classmate, dating the notes with the date of the missed class. When the notebook is graded, the instructor will look for class notes for all absences. As you work through the course you may find some problems require multiple days of work. You may want to have a log where you keep track of dates that you attempt a specific problem. This will allow you to quickly find all attempts at problems as you seek solutions.

2. Each student must also keep a neat and well organized record of all problems and theorems worked on throughout the course. The workbook you are now reading will be this record. Please note: It is very important that problems and theorems **NOT BE ENTERED** into this workbook until **AFTER** being presented in class. Thus, after a problem is presented in class, and you understand the problem, write a rough draft of your solution in your notebook from requirement 1. Then complete the problem in this workbook in a neat and well organized fashion. Think of it this way: Most of you would not turn in a rough draft of an essay as the final product in an English class. Rather, you would polish the essay into a neat and well organized articulation of your ideas. Mathematics should not be any different. This workbook will be collected and graded twice during the semester. Neatness is very important!

The class will begin by essentially developing trigonometry from scratch. Therefore, as you work through each problem and theorem you must not use anything not already discussed in the course, with the exception of a few basic ideas listed in the next paragraph. In particular, you may not use prior trigonometry knowledge before it is developed in the course. Please resist the urge to find solutions on-line.

There are a few ideas it is assumed you are familiar with and you should feel free to use these ideas freely as needed. A few examples:

1. x, y coordinate plane, points, etc.
2. Lines, line segments, rays, etc.
3. Circles, arcs, center, circumference, diameter, radius and tangent lines for circles.
4. Triangles and squares including area formulas.

All completed work must be yours. You may discuss problems with classmates or the instructor, but should not use any other resources. Please avoid looking on-line for solutions to problems or theorems. When you struggle with a problem, discuss the problem with a classmate or the instructor. I emphasize: The goal of this course is for each student to present clear and concise mathematical ideas both in class and in written form in the workbook.

Student grades will be based upon the following items:

1. 100 points: Notebook with class notes and development of mathematical ideas.
2. 300 points: Workbook with neat, well organized solutions to problems and theorems.
3. 100 points: Essays (5 at 20 points each).
4. 100 points: Classroom presentations.
5. 300 points: Quizzes (12 at 25 points each).
6. 300 points: Exams (2 at 150 points each).
7. 200 points: Final Exam.
8. **1,400 Total Points.**

Final grade based on number of total points earned during the course:

1. A: 1260 – 1400 points.
2. B: 1120 – 1259 points.
3. C: 980 – 1119 points.
4. D: 840 – 979 points.
5. F: < 840 points.

Students will be asked to write five short (2+ pages) essays during the semester. Essays should be typed, double space, no more than 12 point font with usual one inch margins and printed on single side. All sources should be cited at the end of the essay. Essays will not be graded for grammatical issues, but ideas should be expressed clearly. Due dates for each essay will be announced in class. Students will always be given at least one week notice of due date.

Chapter 1

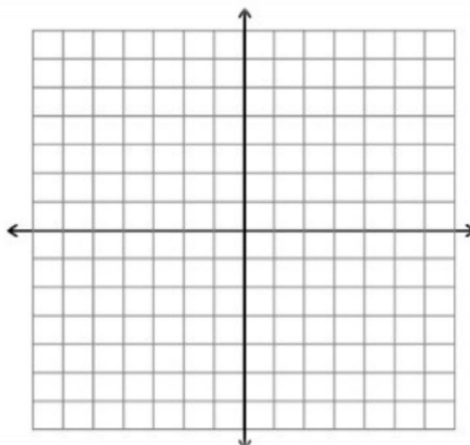
Preliminaries

1.1 Graphing Equations: A Few Basics

Definition 1. ¹ For any real number a , the **absolute value of a** is the positive distance the number a is from zero. The absolute value of a is denoted as $|a|$.

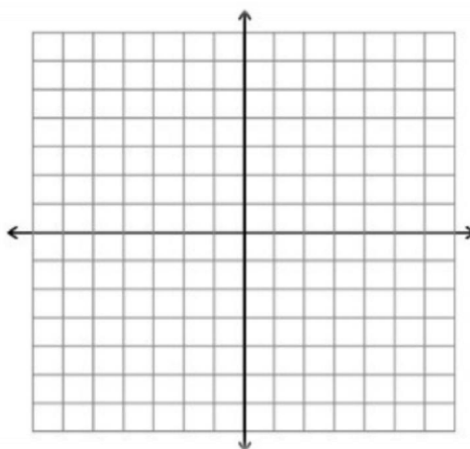
Problem 1. Create a table of values and a graph for $y = |x|$. Give coordinates of x and y intercepts.

x	y



Problem 2. Let $y = |x|$. Replace x with $x - 2$ creating $y = |x - 2|$. Create a table of values and a graph for $y = |x - 2|$. Write a short statement describing what happens to the graph of $y = |x|$ when x is replaced with $x - 2$.

x	y



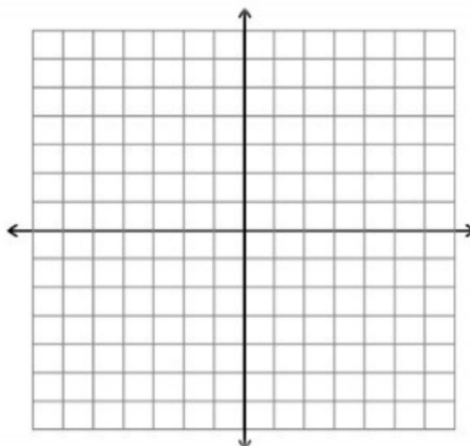
Problem 3. Use graph from problem 1 to solve $|x| \leq 1$.

Problem 4. Solve $|x| \leq 2$.

Problem 5. Explain why the inequality $|x - 3| \leq .01$ means that x must be within .01 units of 3. Using absolute value notation, write an inequality statement expressing that x must be within .005 units of 6. Give a real-world setting where this last statement has practical meaning.

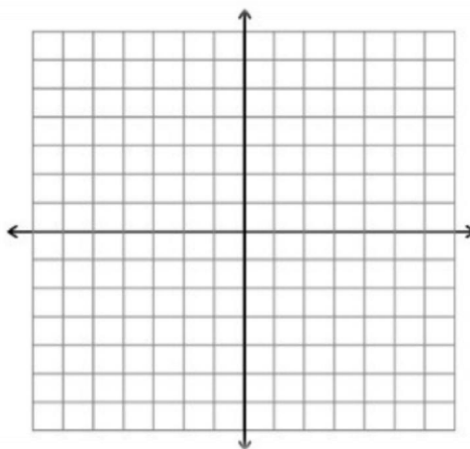
Problem 6. Let $y = |x|$. Replace y with $y - 2$ creating a new equation, $y - 2 = |x|$. Solve this equation for y . Create a table of values and a graph for $y = |x| + 2$. Write a short statement about what you notice.

x	y



Problem 7. Let $y = |x|$. Replace y with $y - 1$ and replace x with $x - 2$. You should get $y - 1 = |x - 2|$. Make a prediction for the graph of $y = |x - 2| + 1$. Create a table and a graph.

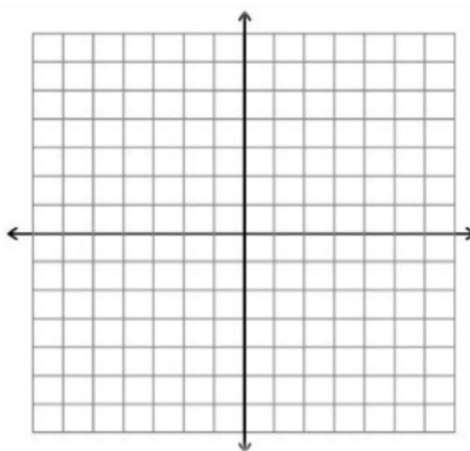
x	y



Definition 2. Let a be any real number and let n be any positive integer. The expression a^n means $a \times a \times \cdots \times a$ where a is multiplied by itself n times. In particular, for any real number a , **the square of a** means $a \times a$ and is denoted as a^2 .

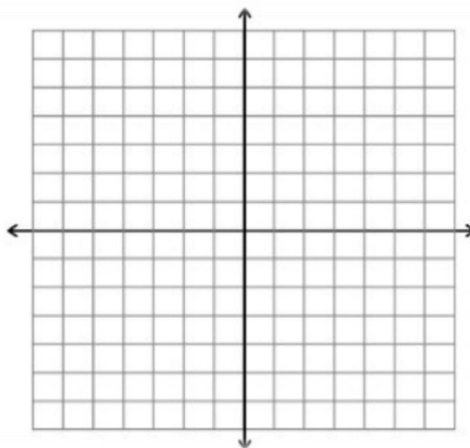
Problem 8. Create a table and a graph of the equation $y = x^2$.

x	y



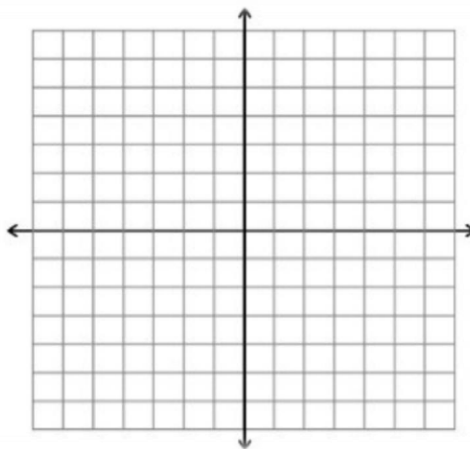
Problem 9. Let $y = x^2$. Replace x with $x + 1$. You should get $y = (x + 1)^2$. Make a prediction for the graph of $y = (x + 1)^2$. Create a table and graph.

x	y



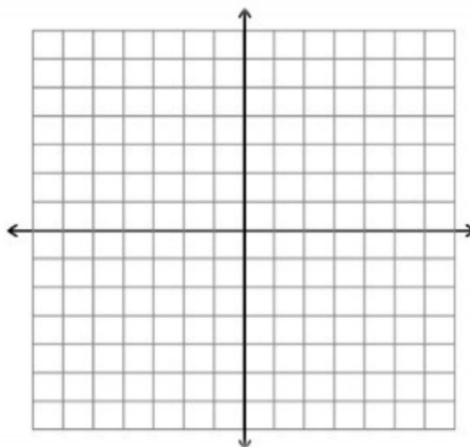
Problem 10. Let $y = x^2$. Replace y with $y - 2$. Make a prediction for the graph of $y = x^2 + 2$. Create a table and graph.

x	y



Problem 11. Let $y = x^2$. Replace y with $y + 1$ and replace x with $x + 3$. Make a prediction for the graph of $y = (x + 3)^2 - 1$. Create a table and graph.

x	y



Definition 3. A **right triangle** is a triangle that has one angle with angle measure 90° .

Definition 4. For any right triangle in a plane, the side opposite the right angle is called the **hypotenuse of the triangle**.

Theorem 1. ² Given a right triangle in a plane with sides of lengths a , b and c where c is the length of the hypotenuse, then $a^2 + b^2 = c^2$.

This is the well-known Pythagorean Theorem.

Theorem 2. *Given any two points in the x,y coordinate plane, (x_1, y_1) and (x_2, y_2) , then the distance between the two points is given by:*

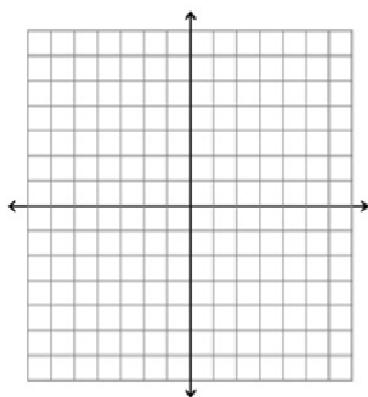
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Chapter 2

Trigonometry

2.1 Radian Measure of an Angle

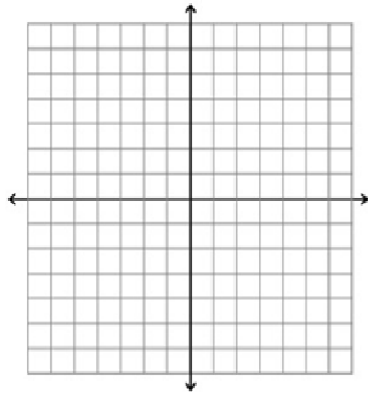
Problem 12. *Graph a circle of radius 1 unit centered at the origin. Use the Pythagorean Theorem to determine an equation relating x and y such that if (x, y) is a point on the circle, then the x, y pair satisfies the equation.*



Definition 5. *The **unit circle** is the circle centered at the origin with radius one unit. You just developed an equation for the unit circle:*

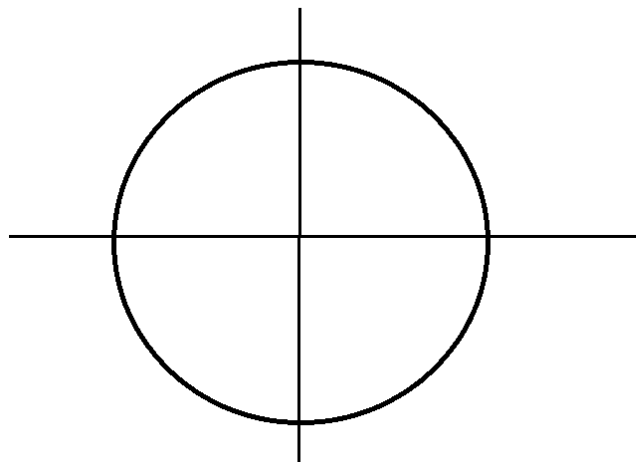
$$x^2 + y^2 = 1$$

Problem 13. In the equation for the circle developed in problem 12, replace y with $y - 3$ and replace x with $x - 3$. Draw a circle for what you expect to be a graph representing $(x - 3)^2 + (y - 3)^2 = 1$. Find two points on the circle and verify the points satisfy the equation.

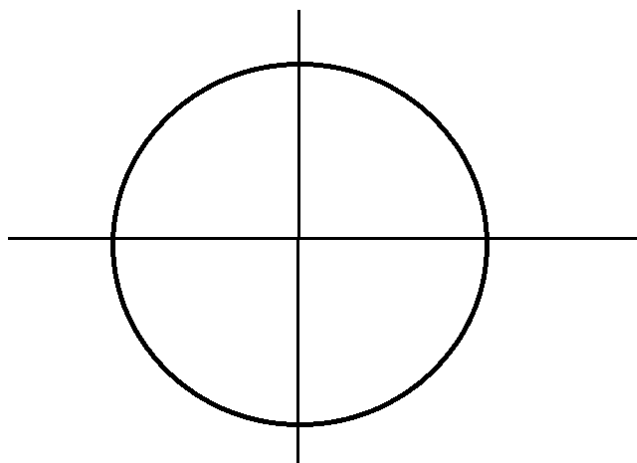


Problem 14. Develop an equation for a circle centered at (h, k) with radius r units.

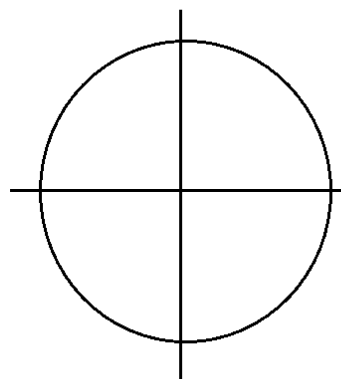
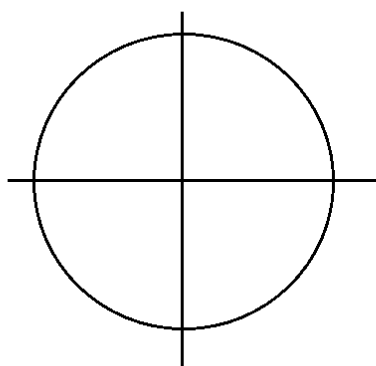
Problem 15. Consider the unit circle placed onto the x, y coordinate plane with the center of the circle placed at the origin. Plot the point $(1, 0)$. Starting at the point $(1, 0)$ and working counter-clockwise, place eight marks on the unit circle that sub-divide the circle into eight equal arcs. Label each point with its arc length distance from the point $(1, 0)$.



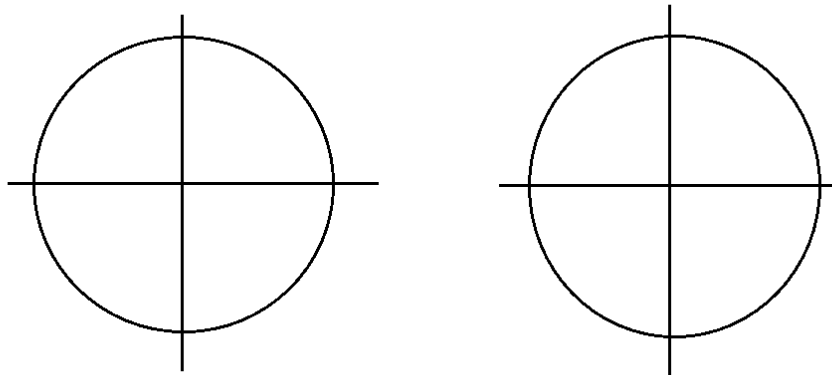
Problem 16. Repeat problem 15 using twelve points.



Problem 17. Repeat problems 15 and 16 using circles centered at the origin with radius 2 units.

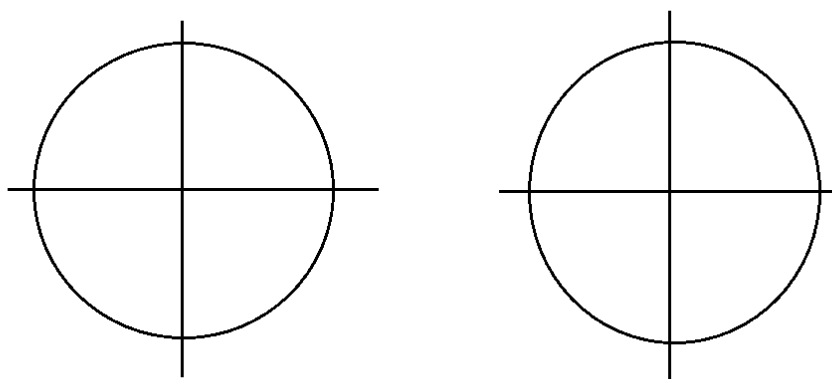


Problem 18. Repeat problems 15 and 16 using circles centered at the origin with radius 3 units.

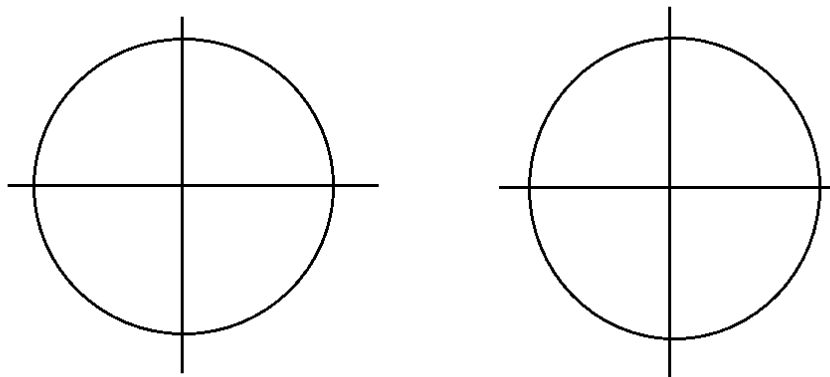


Problem 19. Consider the unit circles from problems 15 and 16. Now label the points with the following ratio:

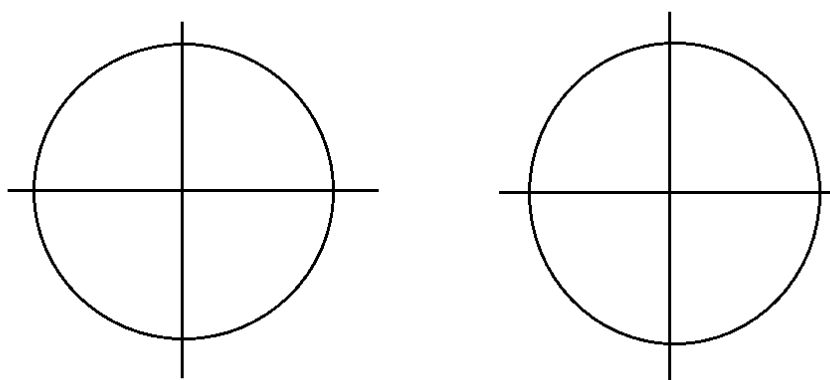
$$\frac{\text{arclength}}{\text{radius}}$$



Problem 20. Repeat problem 19 using circles with radius 2 units.



Problem 21. Repeat problem 19 using circles with radius 3 units.



Definition 6. An *angle* is a subset of a plane consisting of two distinct rays with a common endpoint called the *vertex of the angle*.

Definition 7. An *angle in standard position* is an angle for which the vertex is at the origin and one of the rays is the positive x – axis. The ray on the positive x – axis is referred to as the *initial ray of the angle*. The other ray is referred to as the *terminal side of the angle*.

Definition 8. Given any circle centered at the origin and an angle in standard position, let P_1 be the intersection of the circle with the initial side of the angle and let P_2 be the intersection of the circle and the terminal side of the angle. The **arc associated with the angle** is the portion of the circle traced by a point traversing the circle in a counter-clockwise direction from the point P_1 to the point P_2 .

Definition 9. Given any circle centered at the origin and any angle in standard position, the **radian measure of the angle** is the ratio of the length of the arc associated with the angle and the radius of the circle.

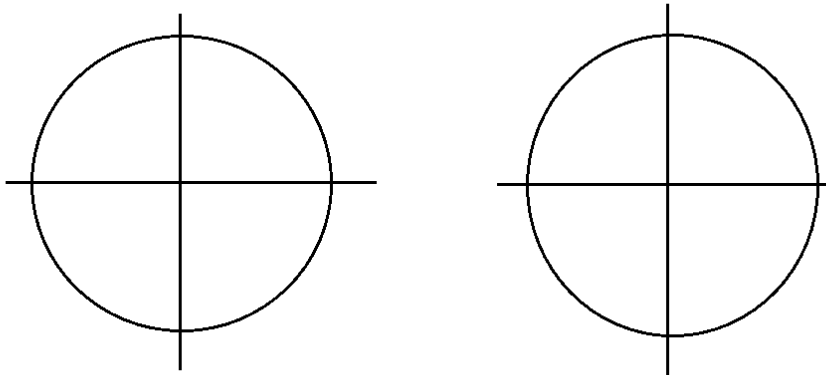
$$\text{radian measure} = \frac{\text{arc length}}{\text{radius}}$$

Notation: If the radian measure of an angle is θ , then we say the angle has radian measure θ .

Problem 22. Draw an angle in standard position where the ray forming the initial side begins at $(0,0)$ and extends through the point $(1,0)$ and the ray forming the terminal side of the angle begins at the point $(0,0)$ and passes through the point $(1,1)$.

Draw a second angle in standard position where the ray forming the initial side begins at $(0,0)$ and extends through the point $(1,0)$ and the ray forming the terminal side of the angle begins at the point $(0,0)$ and passes through the point $(2,2)$.

Find the radian measure for each of these angles.



Definition 10. *The **degree measure of an angle** is $\frac{360}{2\pi}$ times the radian measure of the angle.*

Theorem 3. ³ *The sum of the degree measures of the interior angles of a triangle in a plane is 180° .*

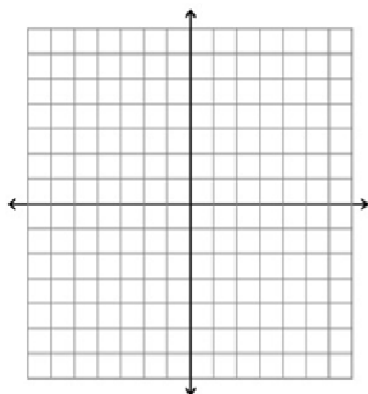
2.2 The Trigonometric Functions

Definition 11. *Two triangles are called **similar triangles** if corresponding angles have equal measure.*

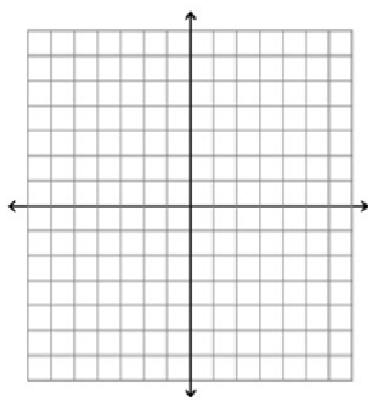
Theorem 4. ⁴ *Given two similar right triangles, then the ratios of corresponding sides are equal.*

Problem 23. *Determine the degree measure of the two angles in problem 22.*

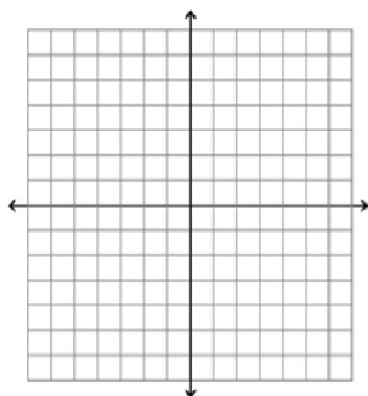
Problem 24. *Draw the unit circle centered at the origin. Draw an angle in standard position that has angle measure 150 degrees. What is the radian measure of this angle?*



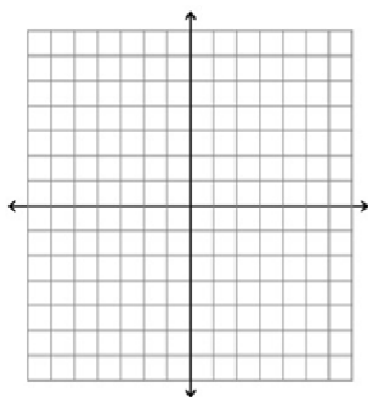
Problem 25. Graph the unit circle centered at the origin. Draw an angle in standard position that has angle measure $\frac{\pi}{4}$ radians. What are the (x,y) coordinates of the intersection of the unit circle and the terminal side of the angle?



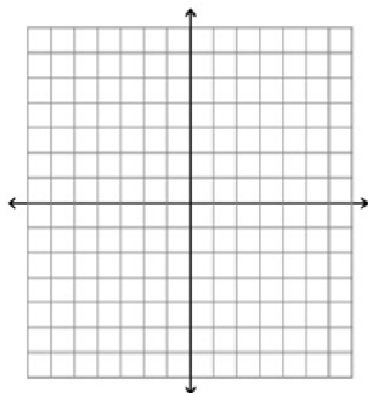
Problem 26. Graph the unit circle centered at the origin. Draw an angle in standard position that has angle measure $\frac{\pi}{6}$ radians. What are the (x,y) coordinates of the intersection of the circle with the terminal side of the angle? What is the length of the arc associated with this angle?



Problem 27. Graph a circle of radius 2 units centered at the origin. Draw an angle in standard position so that the length of the arc associated with the angle is $\frac{3\pi}{2}$ units. What is the radian measure of this angle? What is the degree measure?



Problem 28. Graph a circle of radius 5 units centered at the origin. Draw an angle in standard position that has angle measure $\frac{7\pi}{4}$ radians. Determine the length of the arc associated with this angle.



Problem 29. ⁵ Suppose that a unicycle with a wheel of radius 9 inches is rolled 4 feet. Through what radian measure has one spoke on this wheel traveled? How many revolutions has the wheel made?

Problem 30. Consider the unicycle wheel of problem 29. Suppose it takes 2 seconds to roll the unicycle 2 feet. What is the speed of the unicycle measured in feet/second? What is the speed in mph? What is the angular speed of the unicycle measured in radians/second? How fast is the wheel spinning measured in revolutions/second?

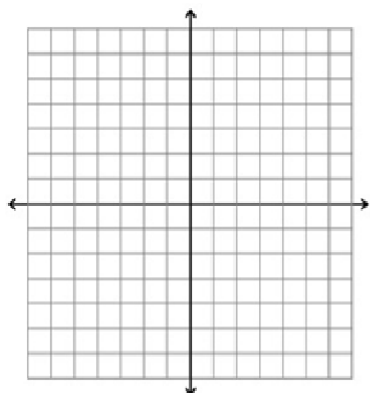
Problem 31. *Given an angle with measure 270° , what is the measure of the angle in radians?*

Problem 32. *Given an angle with angle measure $\frac{2\pi}{3}$ radians, what is the measure of the angle in degrees?*

Notation: ⁶

The development above is dependent on the choice of a counter-clockwise direction in the definition of the arc associated with the angle. An analogous series of definitions could be developed using a clockwise direction. From this point forward we will stay with the convention that an angle measured with a clockwise direction from the positive x – axis will have negative measure and will be referred to as a negative angle while an angle measured with a counter-clockwise direction from the positive x – axis will have a positive measure and will be referred to as a positive angle.

Problem 33. Locate the point on the unit circle so that the angle formed by the radius of the circle containing this point and the positive x – axis has radian measure $-\frac{3\pi}{4}$. What is the measure of this angle in degrees?

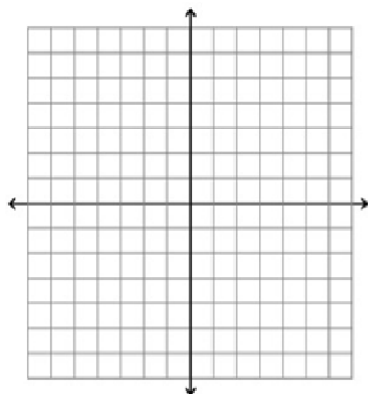


Problem 34. Given an angle with measure -405° , determine the radian measure.

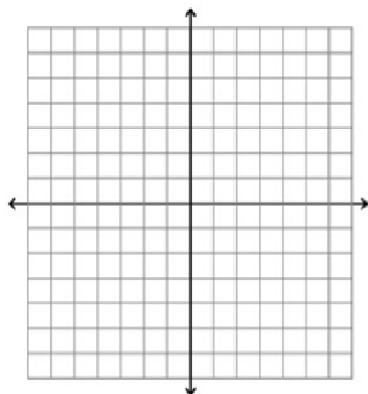
Problem 35. Given an angle with radian measure $-\frac{24\pi}{7}$, determine the degree measure.

Essay 1. Prepare a 2+ page essay addressing the question: Does mathematics have value in our society? Defend your answer and state your sources. Refer to the guidelines for writing an essay given in the introduction of this workbook. The due date for this essay will be given in class.

Problem 36. Determine the coordinates in the (x,y) plane for each point on the unit circle whose distance from the point $(1,0)$ along the unit circle in a counter-clockwise direction is a positive integer multiple of $\frac{\pi}{4}$.



Problem 37. Determine the coordinates in the (x,y) plane for each point on the unit circle such that the angle that is in standard position with the terminal side containing the point has radian measure of a positive integer multiple of $\frac{\pi}{6}$.



Definition 12. ⁷ A **function** is a collection of points in the (x,y) plane with the property that no two of these points are on the same vertical line.

Definition 13. The set of all first coordinates (x - coordinates) of all ordered pairs of a function is referred to as the **domain of the function**. The set of all second coordinates (y - coordinates) of all ordered pairs of a function is referred to as the **range of the function**.

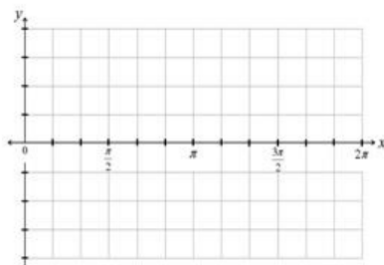
Definition 14. We define the function C such that if $P = (x,y)$ is a point on the unit circle such that the radius of the circle that contains P forms an angle of radian measure θ with the positive x – axis, then $C(\theta) = x$. Hence, $(\theta, x) = (\theta, C(\theta))$ is a point of the function C .

Definition 15. We define the function S such that if $P = (x,y)$ is a point on the unit circle such that the radius of the circle that contains P forms an angle of radian measure θ with the positive x – axis, then $S(\theta) = y$.

Definition 16. We define the function T such that if $P = (x,y)$ is a point on the unit circle such that the radius of the circle that contains P forms an angle of radian measure θ with the positive x – axis, then $T(\theta) = \frac{S(\theta)}{C(\theta)} = \frac{y}{x}$.

Problem 38. Create a table for $y = C(x)$, approximating values to three decimal places. Graph the function $y = C(x)$.

x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	



Problem 39. ⁸

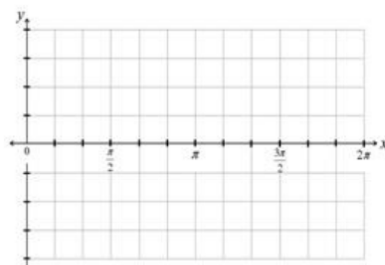
Find a value for θ where $C(\theta) = 0$. Are there other values?

Problem 40. List all values where $C(\theta) = \frac{1}{2}$.

Problem 41. List all values where $C(\theta) = \frac{\sqrt{2}}{2}$.

Problem 42. Create a table and a graph for $y = S(x)$. Approximate values to three decimal places.

x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	



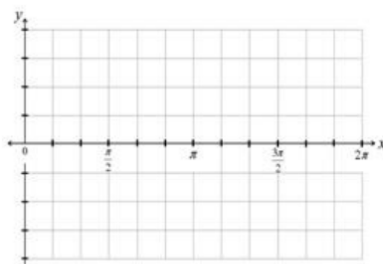
Problem 43. Solve $S(\theta) = 1$ for all possible values of θ .

Problem 44. Solve $S(\theta) = \frac{\sqrt{3}}{2}$ for all possible values of θ .

Problem 45. Solve $S(\theta) = -\frac{1}{2}$ for all possible values of θ .

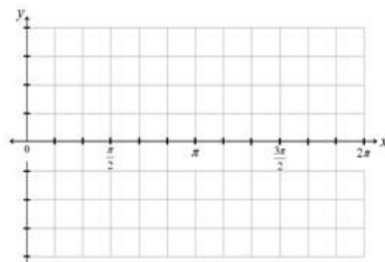
Problem 46. Let $f(x) = -S(x)$ for every real number x . Graph $y = f(x)$.

x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	



Problem 47. Let $g(x) = C(x - \frac{\pi}{2})$ for every real number x . Graph $y = g(x)$.

x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	

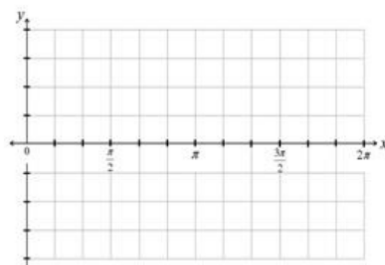


Problem 48. For what values of θ will $T(\theta)$ be undefined? Why?

Problem 49. What is the domain of $T(\theta)$?

Problem 50. Create a table and graph for $y = T(x)$. Approximate values to three decimal places.

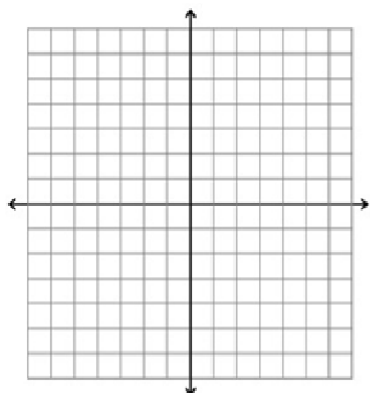
x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	



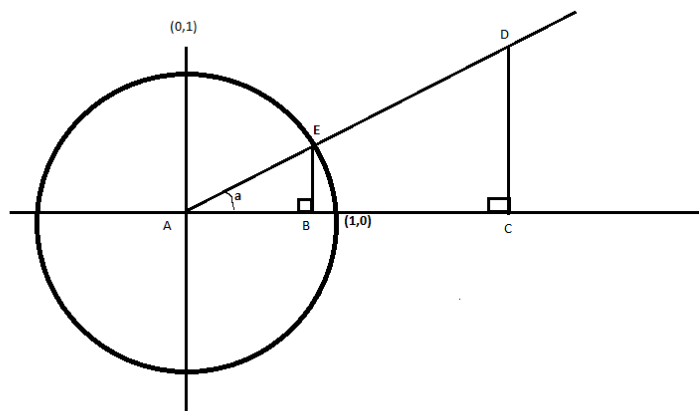
Essay 2. Write a 2+ page essay on any mathematician's contribution to society. Explain any mathematical ideas in terms a classmate could understand.

Problem 51. Solve $T(\theta) = -\frac{\sqrt{3}}{3}$ for all possible values of θ .

Problem 52. Draw the unit circle. Draw a line that is tangent to the circle at the point $(1, 0)$. Let θ be any number with $0 < \theta < \frac{\pi}{2}$. Draw the line that forms an angle with radian measure θ with the positive x – axis. Find the distance from the point $(1, 0)$ to the point of intersection of the two lines you drew.



Note: Refer to the following figure for the next three problems. Also, it will be very helpful to refer to Theorem 4. Notice once again the importance of the Pythagorean Theorem!



Problem 53. Show that $S(a) = \frac{l(CD)}{l(AD)}$.

Problem 54. Show that $C(a) = \frac{l(AC)}{l(AD)}$.

Problem 55. Show that $T(a) = \frac{l(CD)}{l(AC)}$.

Note: We have now defined three functions, C , S and T . The last few problems have shown that if we have a right triangle in a plane with side adjacent to the angle having length a , side opposite the angle having length b and hypotenuse having length c , then these functions satisfy:

$$S(\theta) = \frac{b}{c}$$

$$C(\theta) = \frac{a}{c}$$

$$T(\theta) = \frac{b}{a}$$

Of course these are the trigonometric functions which some of you are familiar with, sine, cosine and tangent, usually abbreviated with $\sin$, $\cos$, $\tan$.

The inputs are always angle measures.

Definition 17. We now define three more functions as reciprocals of the sine, cosine and tangent functions.

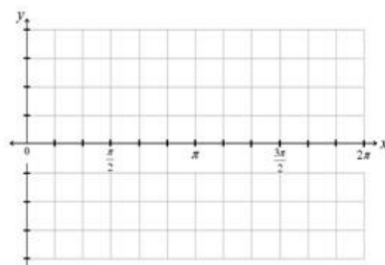
Reciprocal of sine is **cosecant**, $\csc$.

Reciprocal of cosine is **secant**, $\sec$.

Reciprocal of tangent is **cotangent**, $\cot$.

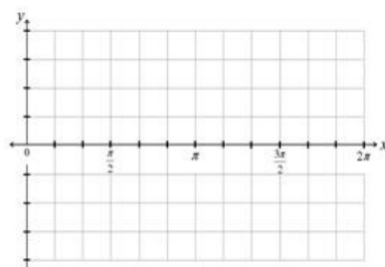
Problem 56. Create a table and a graph for the secant function, $y = \sec(x)$. Give domain and range with an explanation.

x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	



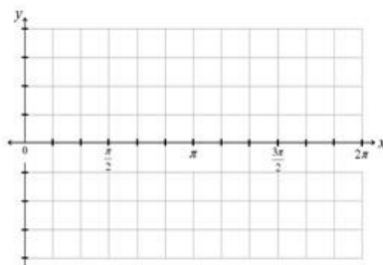
Problem 57. Create a table and a graph for the cosecant function, $y = \csc(x)$. Give the domain and range with an explanation.

x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	



Problem 58. Create a table and graph for the cotangent function, $y = \cot(x)$. Give the domain and range with an explanation.

x	y
0	
$\pi/6$	
$\pi/4$	
$\pi/3$	
$\pi/2$	
$2\pi/3$	
$3\pi/4$	
$5\pi/6$	
π	
$7\pi/6$	
$5\pi/4$	
$4\pi/3$	
$3\pi/2$	
$5\pi/3$	
$7\pi/4$	
$11\pi/6$	
2π	



Definition 18. ⁹ Let $f(x)$ be a function and let A be a positive number. We say the function $f(x)$ has period A if A is the smallest number satisfying $f(x + A) = f(x)$ for all x in the domain of $f(x)$.

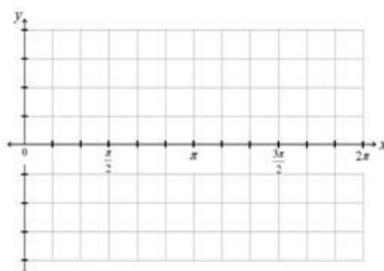
Problem 59. Determine the period for each of the six trigonometric functions.

Problem 60. Create a table and a graph for $y = 3 \sin(x - \pi)$.

Problem 61. Create a table and a graph for $y = \sin(x) + \cos(x)$.

Problem 62. Create a table and graph for $y = x \sin(x)$.

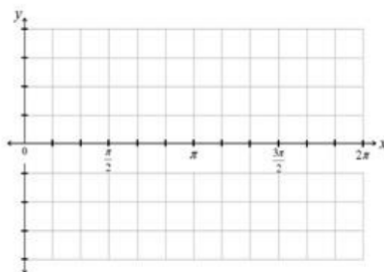
x	y



Problem 63. Consider $y = 3 \sin(2x - \frac{\pi}{6})$. First, write a sentence describing in detail how this is a transformation of $y = \sin(x)$. Create a table and graph. Include two cycles in your graph.

Sentence:

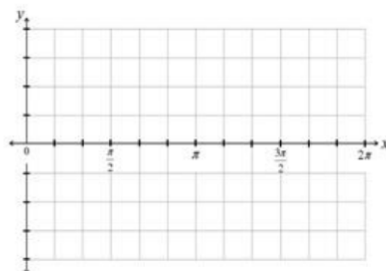
x	y



Problem 64. Consider $y = 3 \sin(3x - \frac{\pi}{18})$. First, write a sentence describing in detail how this is a transformation of $y = \sin(x)$. Create a table and graph. Include two cycles in your graph.

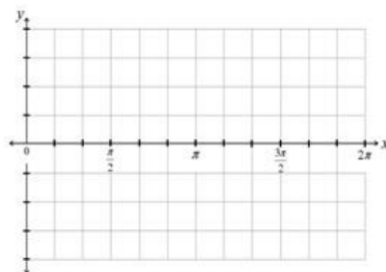
Sentence:

x	y



Problem 65. Create a table and graph for $y = 5 - \cos(2x)$.

x	y



2.3 Trigonometric Equations

Problem 66. *At what minimum height above ground level must a satellite dish be placed so that at a 30° angle it will be able to see over a building that is 40 feet tall and 50 feet away from the satellite dish?*

Problem 67. *Solve $2\sin(x) = 1$ for all possible values of x .*

Problem 68. *Solve $2\cos(\theta) = -\sqrt{3}$ for all possible values for θ .*

Problem 69. Solve $\tan(\theta) = 1$ for all possible values for θ .

Problem 70. Solve $\cos^2(x) - 1 = 0$ for all possible values for x .

Problem 71. Solve $\sin^2(x) - \sin(x) = 0$ for all possible values for x .

Problem 72. Solve $2\sin^2(x) - 3\sin(x) + 1 = 0$ for all possible values for x .

Problem 73. Solve $\cot(x) > 0$ for all possible values for x .

Problem 74. Solve $\tan(x) < 0$ for all possible values for x .

Essay 3. Write a 2+ page essay describing how you will use mathematics after you graduate from college.

Problem 75. ¹⁰ Let S be a square. Let M and N be the midpoints on two adjacent sides. Let V be the corner point that is opposite both points M and N . Let θ be the angle measure of the angle formed by two lines connecting the point V with the points M and N . Compute $\sin(\theta)$.

Problem 76. Solve $\csc(x) = -\frac{2\sqrt{3}}{3}$ for all possible values for x .

Problem 77. Create a table and graph for $y = -2 + 4\cos(\pi x + \frac{\pi}{4})$.

Problem 78. Solve $2\sin(4x) - \sqrt{3} = 0$ for all possible values for x .

Problem 79. *A crop duster plane passes directly over your head at an altitude of 500 feet. Two seconds later you observe the plane with the same altitude and an angle of elevation of 42° from ground level. Find the plane's average speed over the two seconds.*

Problem 80. *Solve $2\sin\left(\frac{x}{3}\right) + 1 = 0$ for all possible values for x .*

Problem 81. Solve $\cos^2(2x) - 2\cos(2x) + 1 = 0$ for all possible values of x .

Problem 82. Assume θ is any real number such that neither $\sin(\theta)$ nor $\cos(\theta)$ is zero. Simplify $\tan(\theta)\cot(\theta)$. Explain why we made the assumption. To simplify means to write the trigonometric expression using fewer algebraic operations.

2.4 Trigonometric Identities

Problem 83. *Show that $\sin^2(x) + \cos^2(x) = 1$ for all values of x .*

Problem 84. *Show that $\tan^2(x) + 1 = \sec^2(x)$ for all values of x .*

Problem 85. *Show that $\cot^2(x) + 1 = \csc^2(x)$ for all values of x .*

Notation: The three identities above are called the Pythagorean Identities.

Problem 86. ¹¹ Solve $1 + \sin(x) = \cos(x)$ for all possible values of x .

Problem 87. Solve $\csc(x) + \cot(x) = 1$ for all possible values of x .

Problem 88. Solve $\cos^2(x) = \cos(x) + \sin^2(x)$ for all possible values of x .

Problem 89. Simplify $\frac{\tan(x)}{(\sec(x)\csc(x))}$. For what values of x is this expression defined? For what values of x is your simplified expression defined? Are the two expressions defined for all values of x ?

Problem 90. Simplify $(\cos^2(x) - 1)(\tan^2(x) + 1)$. For what values of x is this expression meaningful? For what values of x is your simplified expression meaningful? Are the two expressions equal for all values of x ?

Problem 91. Simplify $\tan(x)\cos(x)$.

Problem 92. Simplify $\frac{\sec(x)}{\csc(x)}$.

Problem 93. Simplify $\frac{\sec(x) + \csc(x)}{1 + \tan(x)}$.

Problem 94. Simplify $(\csc(x) + 1)(\csc(x) - 1)$.

Problem 95. *Simplify* $\frac{1+\sec(x)}{\tan(x)+\sin(x)}$.

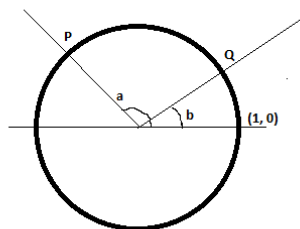


Figure 1

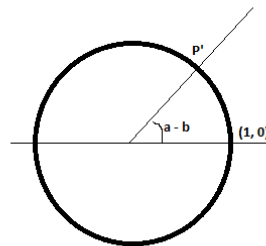


Figure 2

Problem 96. For the unit circle in figure 1, find the distance from P to Q in terms of $\sin(a)$, $\cos(a)$, $\sin(b)$ and $\cos(b)$.

Problem 97. On the unit circle in figure 2, find the distance from P' to $(1, 0)$ in terms of $\sin(a - b)$ and $\cos(a - b)$.

Problem 98. Give a justification why the distances found in problems 96 and 97 are equal. Set the distances equal to each other and simplify.

Note: The identity found in problem 98 is called the cosine difference identity. This identity leads to a variety of identities which will now be developed.

Definition 19. If $f(x)$ is a function and $f(-x) = f(x)$ for every x in the domain of $f(x)$, then we say that $f(x)$ is **an even function**.

Definition 20. If $f(x)$ is a function and $f(-x) = -f(x)$ for every x in the domain of $f(x)$, then we say that $f(x)$ is **an odd function**.

Problem 99. Let $f(x)$ be any function such that if a is in the domain of $f(x)$, then $-a$ is also in the domain of $f(x)$. Show that $f(x)$ can be written as the sum of an even function and an odd function.

Note: Even and odd functions correspond to symmetry about the y – axis and symmetry about the origin respectively.

Problem 100. Let $f(x) = x^2$, $g(x) = x^3$ and $h(x) = f(x) + g(x)$. Show that $f(x)$ is an even function, $g(x)$ is an odd function and $h(x)$ is neither an even function nor an odd function.

Problem 101. Determine which of the six trigonometric functions are even functions and which are odd functions.

Problem 102. *Prove that if $f(x)$ is an even function and $g(x)$ is an odd function, then $h(x) = f(x)g(x)$ is an odd function.*

Problem 103. *Substitute $b = -b$ into the cosine difference identity and develop the cosine sum identity.*

Problem 104. *Substitute $a = \frac{\pi}{2}$ into the cosine difference identity and develop one of the co-function identities (translation identity). What does this co-function identity say about the graphs of $\sin(x)$ and $\cos(x)$? List at least three additional co-function identities.*

Problem 105. *Substitute $b = b - \frac{\pi}{2}$ into the cosine sum identity and develop another identity. What would you call this identity?*

Problem 106. *There are four difference and sum identities for cosine and sine functions. Which three have been developed? Develop the fourth.*

Problem 107. *Develop the tangent difference identity by simplifying the quotient of the sine difference identity and the cosine difference identity.*

Problem 108. *Develop the tangent sum identity.*

Problem 109. *Prove or disprove: $\cot(x) + \cot(y) = \frac{\cos(x-y)}{\cos(x)\sin(y)}$.*

Problem 110. *Use identities to prove $\cot(\frac{\pi}{2} - x) = \tan(x)$.*

Problem 111. *Compute an exact value for $\cos(\frac{\pi}{12})$ and use this to construct a right triangle to compute all trigonometric values for both $\frac{\pi}{12}$ and $\frac{5\pi}{12}$.*

Problem 112. *Prove $\sin(2x) = 2\sin(x)\cos(x)$ is valid for all values of x .*

Problem 113. *Prove $\cos(2x) = \cos^2(x) - \sin^2(x)$ is valid for all values of x . Use a Pythagorean Identity to develop two other identities for $\cos(2x)$ that are valid for all values of x .*

Problem 114. *Prove $\tan(2x) = \frac{2\tan(x)}{1-\tan^2(x)}$ is valid for all values of x for which both sides of the identity are defined.*

Note: The last few problems have developed identities that are referred to as the double angle identities. They are quite useful in calculus. The next two are referred to as the half angle identities and can be developed quite easily from the double angle identities once you see what substitutions to make.

Problem 115. Prove $\sin^2\left(\frac{a}{2}\right) = \frac{1-\cos(a)}{2}$ is valid for all values of a .

Problem 116. Prove $\cos^2\left(\frac{a}{2}\right) = \frac{1+\cos(a)}{2}$ is valid for all values of a .

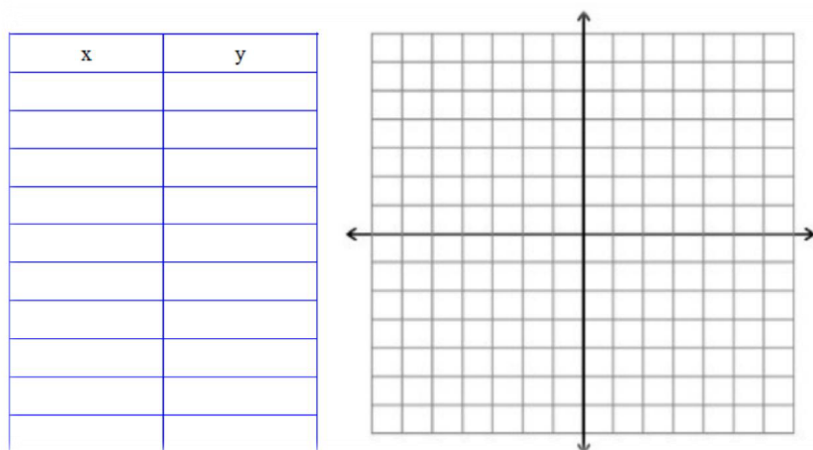
Problem 117. Solve $\sin(2x) = 2\sin(x)\cos(x)$ for all possible values of x .

2.5 Inverse Trigonometric Functions

Note: Recall that a function is a collection of ordered pairs satisfying a certain property. Not every function has an inverse function, but if one does, then the inverse of the function is the collection of ordered pairs obtained by reversing the 1st and 2nd coordinates of the function. Thus if $f(x)$ has an inverse function, then we denote the inverse function as $f^{-1}(x)$. $f^{-1}(x)$ is the inverse of $f(x)$ if and only if the functions satisfy $f^{-1}(x) = y$ and $f(y) = x$. One difficulty that sometimes occurs is when the coordinates of a function are reversed, the new collection of ordered pairs may not satisfy the requirements of a function. Namely, in the new collection, there may be multiple ordered pairs where a single domain value has multiple range values. For example, $f(x) = x^2$ contains the ordered pairs $(1, 1)$ and $(-1, 1)$. When the coordinates are reversed, the new collection of ordered pairs contains $(-1, 1)$ and $(1, 1)$, which is clearly problematic since two domain values have the same range value. Often times, this difficulty can be corrected by restricting the domain of the inverse function. For $f(x) = x^2$ we can say the inverse function is $f^{-1}(x)$ on the domain $0 \leq x < \infty$ and range $0 \leq x < \infty$. The inverse functions for the six trigonometric functions will be defined in a similar way, with restricted domain and range values.

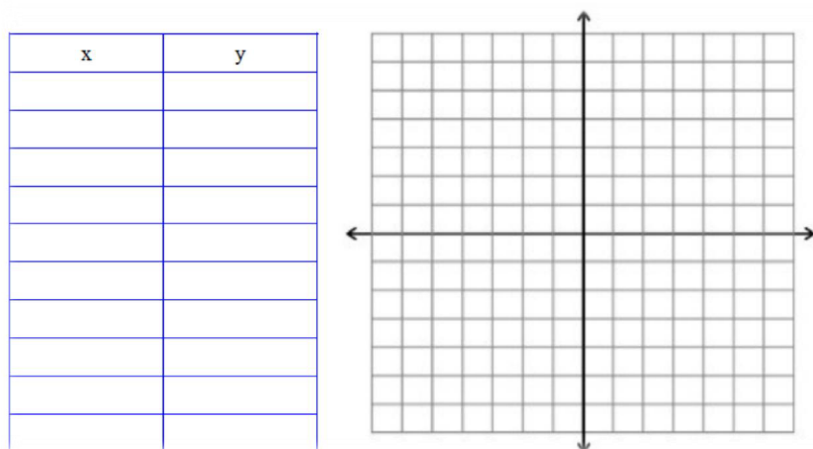
Definition 21. We define the **inverse sine function**, $\sin^{-1}(x)$ or $\arcsin(x)$ to be the function satisfying $\sin^{-1}(x) = y$ if and only if $\sin(y) = x$ and having domain $-1 \leq x \leq 1$ and range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

Problem 118. Create a table and graph for the inverse sine function.



Definition 22. We define **inverse cosine function**, $\cos^{-1}(x)$ or $\arccos(x)$, to be the function satisfying $\cos^{-1}(x) = y$ if and only if $\cos(y) = x$ and having domain $-1 \leq x \leq 1$ and range $0 \leq y \leq \pi$.

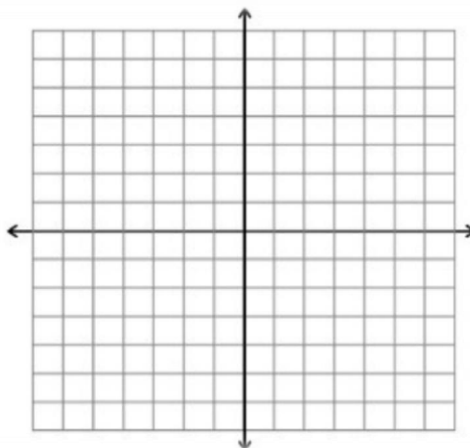
Problem 119. Create a table and graph for the inverse cosine function.



Definition 23. We define the *inverse tangent function*, $\tan^{-1}(x)$ or $\arctan(x)$, to be the function satisfying $\tan^{-1}(x) = y$ if and only if $\tan(y) = x$ and having domain $-\infty < x < \infty$ and range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

Problem 120. Create a table and graph for the inverse tangent function.

x	y



Note: The $\arcsin(x)$, $\arccos(x)$, $\arctan(x)$ functions are often written respectively as $\sin^{-1}(x)$, $\cos^{-1}(x)$, $\tan^{-1}(x)$. This is an abuse of notation since $\sin^{-1}(x) \neq \frac{1}{\sin(x)}$. The convention is that $\sin^n(x) = (\sin(x))^n$ except when $n = -1$. When needed, $\frac{1}{\sin(x)} = (\sin(x))^{-1}$.

Problem 121. What is the value for $\arcsin(\frac{-\sqrt{3}}{2})$?

Problem 122. Compute $\arctan(1)$.

Problem 123. What is the difference between the next two problems? Solve both problems and state the differences.

1. Solve $\cos(\theta) = \frac{1}{2}$.

2. Compute: $\arccos(\frac{1}{2})$.

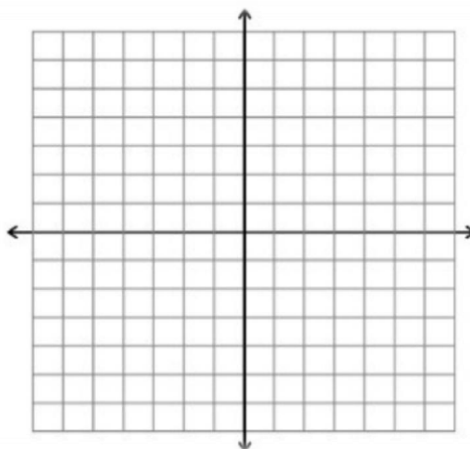
Problem 124. Compute $\arccos(-\frac{\sqrt{2}}{2})$.

Problem 125. Compute $\sin(\arctan(\frac{3}{4}))$.

Problem 126. Compute $\cot(\cos^{-1}(-\frac{2}{3}))$.

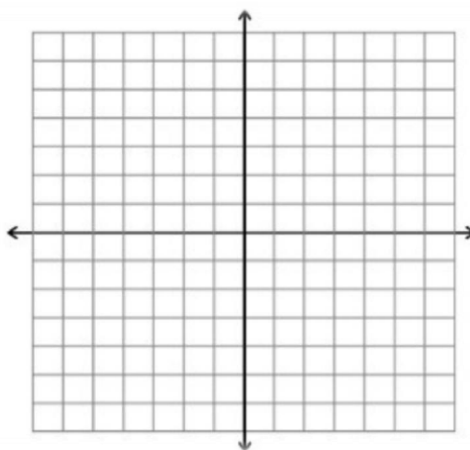
Problem 127. Create a table and graph for $y = \sin^{-1}(x+3)$ for $-4 \leq x \leq -2$.

x	y



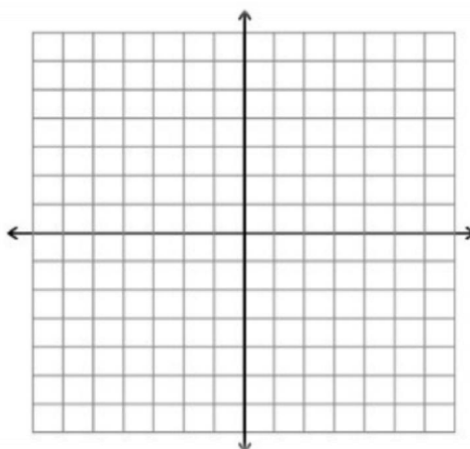
Problem 128. Create a table and graph for $y = -2\cos^{-1}(\frac{x}{3})$ for $-3 \leq x \leq 3$.

x	y



Problem 129. Create a table and graph for $y = \tan^{-1}(1 - 2x)$ for $-2 \leq x \leq 3$.

x	y



Theorem 5. ¹² Prove that the solutions to $ax^2 + bx + c = 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note: This is the well-known quadratic formula.

Problem 130. Solve $\tan^2(x) = 2\tan(x) + 1$ for all values of x . Use your calculator and approximate your answers to four digits.

Problem 131. Solve $\sin^2(x) + 2 = 4\sin(x)$ for all values of x . Use your calculator and approximate your answers to four digits.

2.6 Law of Sines and Law of Cosines

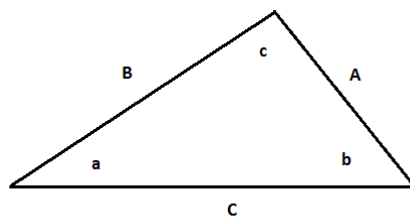


Figure 11

Problem 132. Use figure 11 to show $\frac{A}{\sin(a)} = \frac{B}{\sin(b)}$.

Problem 133. Use figure 11 to show $A^2 + B^2 - 2AB\cos(c) = C^2$.

Note: One very common convention is to label triangle angle measures with α, β and γ and to label triangle side lengths with a, b and c with angle α opposite side a , angle β opposite side b and angle γ opposite side c . Use this convention for the following problems. So solve a particular triangle means to find all side lengths and all angle measures.

Problem 134. Draw and solve any triangle satisfying: $\alpha = 32^\circ$, $\beta = 41^\circ$ and $a = 2.5$ cm.

Problem 135. Draw and solve any triangle satisfying: $\alpha = 29^\circ$, $a = 7$ cm, and $c = 14$ cm.

Problem 136. *Draw and solve any triangle satisfying: $\alpha = 43^\circ$, $b = 6$ cm and $c = 10$ cm.*

Problem 137. *Draw and solve any triangle satisfying: $\beta = 72.2^\circ$, $b = 78.3$ cm, and $c = 145$ cm.*

Problem 138. *Draw and solve any triangle satisfying: $a = 6$ cm, $b = 8$ cm and $c = 9$ cm.*

Problem 139. *Draw and solve any triangle satisfying: $\alpha = 26^\circ$, $a = 11$ cm and $b = 18$ cm.*

Problem 140. *Draw and solve any triangle satisfying: $a = 8$ cm, $\beta = 60^\circ$ and $c = 11$ cm.*

Problem 141. *Draw and solve any triangle satisfying: $\alpha = 20^\circ$, $a = 10$ cm and $b = 16$ cm.*

Problem 142. *Find the length of one side of a pentagon inscribed in a circle of radius 12.5 cm.*

Problem 143. *Find the length of one side of a nine-sided regular polygon inscribed in a circle with a radius of 8.32 cm.*

Problem 144. *Two bird watchers located at points A and B are 12.5 miles apart. A Yellow-Bellied Sap Sucker is seen at point C by both bird watchers. Careful measurements indicate that $m\angle BAC = 14^\circ$ while $m\angle ABC = 82^\circ$. Which bird watcher is closest to the Sap Sucker? How far away is the bird from this bird watcher?*

Problem 145. *An observer stands at a point A at ground level that is some unknown distance from the base of a building. The angle of elevation from point A to the top of the building is 63° . After taking this measurement, the observer walks directly away from the building for 140 feet to a point B also at ground level. The observer now measures the angle of elevation from the point B to the top of the building as 55° . What is the height of the building?*

Notation: To solve the following problems requires a bit of an introduction to navigation. A direction such as $N30^\circ E$ is read as the direction 30° East of true North. In other words, from true North, rotate clockwise 30° to get the direction of travel, also called a heading. As another example, the heading $S40^\circ E$ means from due South, rotate counterclockwise 40° to get direction of travel.

Problem 146. *A sailor spots a lighthouse at a heading of $N28^\circ E$. The sailor then sails due East 7.5 miles where he then spots the lighthouse at a heading of $N16^\circ E$. At this point, how far is the sailor from the lighthouse? Continuing to sail due East, how far must he sail to reach a point closest to the lighthouse?*

Problem 147. *In order to seal an oil pipeline we must make a plate of $\frac{1}{4}$ inch steel to place on a flange at the open end of an open pipe. Our plate must be circular with a radius of 6 inches and must have 7 holes drilled into it spaced equally apart. The holes must have diameter $\frac{3}{16}$ inch and their centers must be 1 inch from the perimeter of the plate. What will be the distance between the centers of two adjacent holes? Industry standards require an answer accurate to 6 decimal places.*

Problem 148. *Jack and Jill take off from the same airport at the same time in their Cessna airplanes. Jack travels $N35^\circ W$ at 160 mph while Jill travels $S70^\circ W$ at 170 mph. How far apart are their planes after 2 hours? Determine a function that will give the distance between the planes after t hours. Use the function to determine the distance after 5 hours.*

Problem 149. *Two joggers in Central Park are resting on park benches at points A and B. Point A is 1.2 miles directly south of point B. At midnight both joggers spot a walker traveling in a straight line path. The jogger resting at point A observes the walker at a heading of $N20^\circ E$, while the jogger resting at point B observes the same walker at a heading of $S70^\circ E$. Ten minutes later the jogger resting at point A observes the walker at a heading of $N34^\circ E$, while the jogger resting at point B observes the walker at a heading of $S55^\circ E$. Find the walker's average walking speed.*

Essay 4. Write a 2+ page essay on Trigonometry.

2.7 Polar Coordinates

Note: Trigonometry gives us another way to coordinate the plane. Algebra has given us the Cartesian Coordinate system which consists of two perpendicular lines representing two real number lines. The intersection of the two lines is called the *origin*. The horizontal line is the x – *axis* and the vertical line is the y – *axis*. From any point in the plane a vertical line is drawn intersecting the x – *axis* at the point's x – *coordinate* and a horizontal line is drawn intersecting the y – *axis* at the point's y – *coordinate*. Thus the point is located by the rectangular coordinates, (x, y) . Now trigonometry gives us what we call the polar coordinates of any point.

Definition 24. *Construct a Cartesian Coordinate system. For any point $p \neq (0, 0)$ in the plane let r be the linear distance from the point p to the origin. Construct a ray from the origin to the point p . This ray forms an angle with the x – *axis*. Let θ be the measure of the angle formed. Then the coordinates (r, θ) are called the **Polar Coordinates** of the point p . The origin is the point $(0, 0)$ in polar coordinates as well as in rectangular coordinates.*

Problem 150. *Establish formulas for r and θ in terms of x and y that will convert any point from rectangular coordinates (x, y) to polar coordinates (r, θ) .*

Problem 151. *Convert the point with rectangular coordinates $(x, y) = (1, 1)$ into polar coordinates (r, θ) .*

Problem 152. *Convert the point with rectangular coordinates $(x, y) = (-2, -5)$ into polar coordinates (r, θ) .*

Problem 153. *Establish a formula for x and y that will convert any point with polar coordinates (r, θ) into rectangular coordinates (x, y) .*

Problem 154. *Convert the point with polar coordinates $(5, 30^\circ)$ into rectangular coordinates (x, y) .*

Problem 155. *Convert the point with polar coordinates $(\frac{8}{3}, \frac{3\pi}{5})$ into rectangular coordinates (x, y) .*

Chapter 3

Conic Sections

In this course we have seen circles and given a particular circle with radius r units and center (h, k) we saw that all points x, y on the circle satisfy the equation:

$$(x - h)^2 + (y - k)^2 = r^2$$

We have also seen parabolas and given a particular parabola with a given value for a and vertex (h, k) we saw that the all points x, y on the parabola satisfy the equation $y - k = a(x - h)^2$. These are two examples of what we will now call conic sections.

Definition 25. *Consider two identical, infinitely tall right circular cones placed vertex to vertex so that they share the same axis of symmetry. A **conic section** is the intersection of this three dimensional surface with any plane that does not pass through the vertex where the two cones intersect.*

Note: These intersections are called circles when the plane is perpendicular to the axis of symmetry, lines when the plane is parallel and tangent to a side of the cone, parabolas when the plane is parallel to a side of the cone, but not tangent, hyperbolas when the plane is parallel to the axis of symmetry and ellipses when the plane does not meet any of the previous criteria.

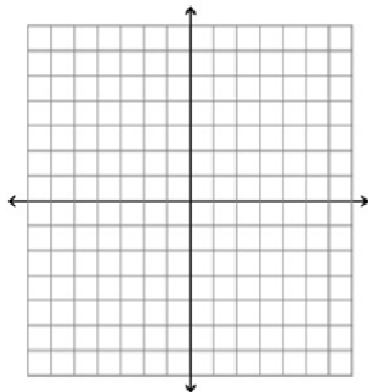
This gives a geometric description of conic sections. Any conic section may also be represented by a quadratic equation in two variables, x and y .

Every conic section may be obtained by making appropriate real value choices for A, B, C, D, E and F in the following equation:

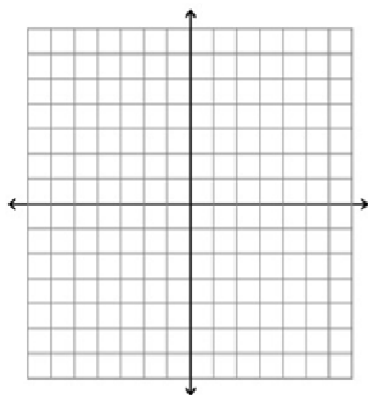
$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

Problem 156. Graph the set of all points (x,y) in the plane satisfying $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ where:

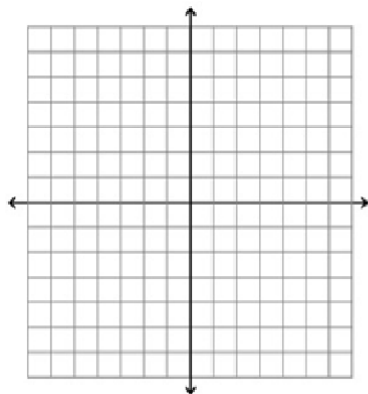
1. $A = B = C = 0, D = 1, E = 2$ and $F = 3$



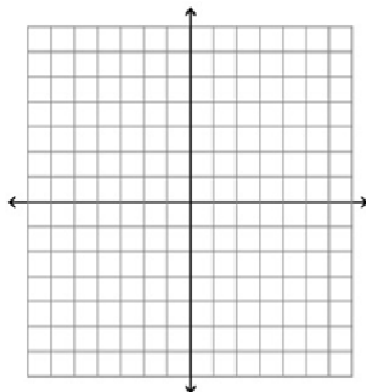
2. $A = \frac{1}{16}, C = \frac{1}{16}, B = D = E = 0$ and $F = -1$



3. $A = B = C = D = E = 0$ and $F = 1$



4. $A = B = C = D = E = F = 0$



Definition 26. Given a plane, a point h, k in the plane and a positive number r , the circle with **center** (h, k) and **radius** r is the set of all points in the plane that are a distance r units from the point (h, k) .

Problem 157. Determine values for A, B, C, D, E and F so that $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ represents a circle. A single point. A parabola.

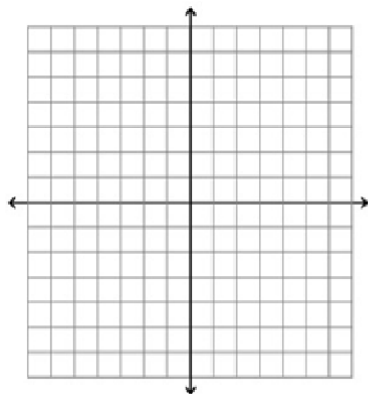
Problem 158. Use the method of completing the square as discussed in class to rewrite the equation $4x^2 + 4y^2 + 6x - 8y - 1 = 0$ in the form of $(x - h)^2 + (y - k)^2 = r^2$ for some h, k and r . Now generalize your work to rewrite the equation $x^2 + y^2 + Dx + Ey + F = 0$ in the same form.

Definition 27. The *distance between a point P and a line L* (denoted by $d(P, L)$) is defined to be the length of the line segment perpendicular to L that has one end on L and the other end on P .

Definition 28. Given a point P and a line L in the same plane, the **parabola** defined by P and L is the set of all points Q in the plane so that $d(P, Q) = d(Q, L)$. The point P is called the **focus** of the parabola. The line L is called the **directrix** of the parabola.

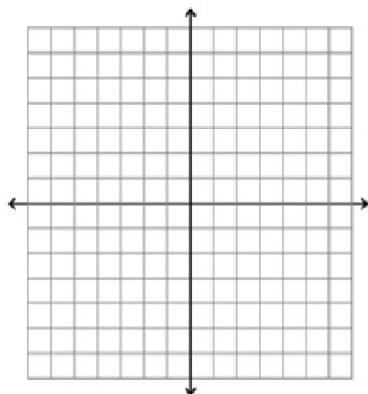
Problem 159. *Parabolas:*

1. Let p be any positive number. Graph the parabola that has focus $(0, p)$ and directrix $y = -p$.



2. Let (x, y) be an arbitrary point on the parabola with focus $(0, p)$ and directrix $y = -p$. Show that (x, y) satisfies the equation $4py = x^2$.
3. Let (x, y) be an arbitrary point on the parabola with focus $(p, 0)$ and directrix $x = -p$. Show that (x, y) satisfies the equation $4px = y^2$.

4. Graph the parabola, focus and directrix for the parabola $y = -x^2 + 3$.



Problem 160. Find the vertex, focus and directrix for each of the parabolas defined by the following equations:

1. $y = 3x^2 - 12x + 27$

2. $y = x^2 + cx + d$ where c and d are any real numbers.

Definition 29. Given any two points, F_1 and F_2 in a plane and a positive number d , the **ellipse** determined by F_1, F_2 and d is the set of all points Q in the plane so that $d(F_1, Q) + d(F_2, Q) = d$. F_1 and F_2 are called the **foci** of the ellipse. The **major axis** of the ellipse is the line passing through the foci points. The **vertex points** of the ellipse are the points where the ellipse intersects with the major axis. The **minor axis** is the line perpendicular to the major axis and passing through the point that is midpoint of the foci

Problem 161. Let $F_1 = (c, 0)$, $F_2 = (-c, 0)$ and $d = 2a$. Show that if (x, y) is a point of the ellipse determined by F_1, F_2 and d , then (x, y) satisfies the equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ where } a^2 - b^2 = c^2$$

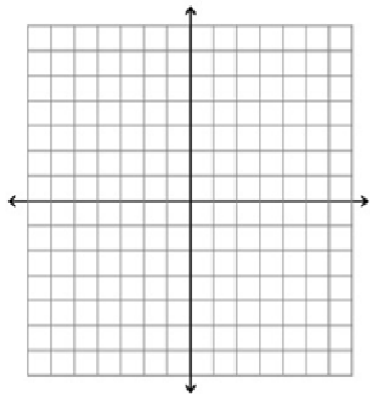
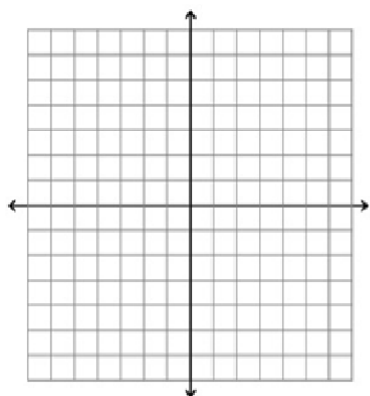
What are the restrictions on a, b and c ?

Problem 162. Ellipses:

Graph the ellipses defined by:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1 \text{ and } \frac{(x+1)^2}{25} + \frac{(y-2)^2}{9} = 1$$

List the foci, vertex points and major axis for each ellipse. Describe any similarities between the two graphs.



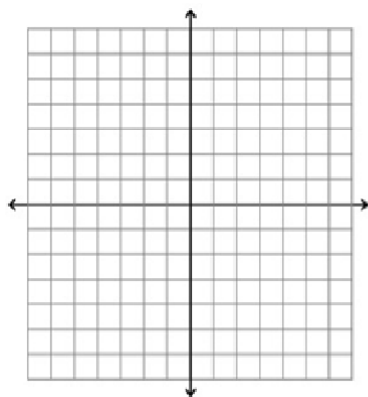
Definition 30. Given any two points F_1 and F_2 in a plane and a positive number d , the **hyperbola** determined by F_1 , F_2 and d is the set of all points Q in the plane so that $|d(F_1, Q) - d(F_2, Q)| = d$. F_1 and F_2 are called the **foci** of the hyperbola.

Problem 163. Let $F_1 = (c, 0)$, $F_2 = (-c, 0)$ and $d = 2a$. Show that the equation of the hyperbola determined by F_1 , F_2 and d is given by:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ where } c^2 = a^2 + b^2.$$

What are the restrictions on a, b and c ?

Problem 164. Graph any hyperbola of your choice. Just choose any real numbers a and b .



Problem 165. Show that as $x \rightarrow \infty$ the hyperbola approaches the lines:

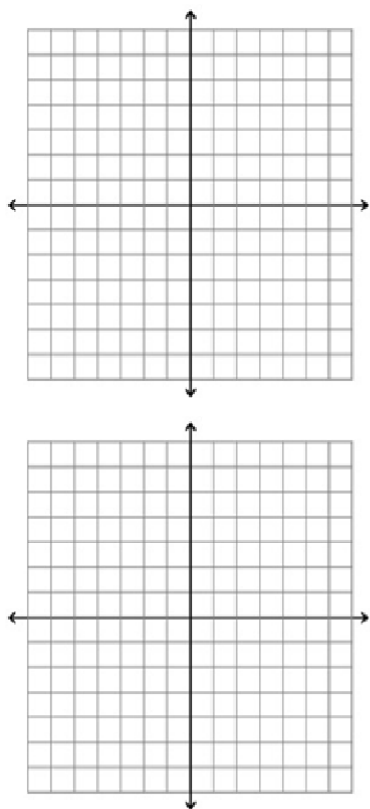
$$L = \pm \frac{b}{a}x$$

Thus the asymptotic behavior of hyperbolas is linear.

Problem 166. Hyperbolas:

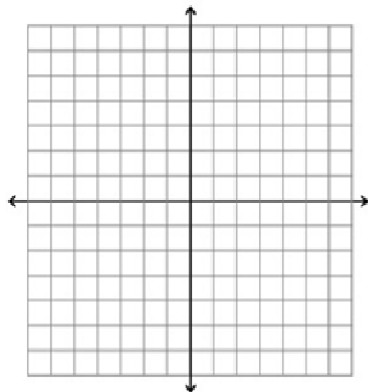
Graph $\frac{x^2}{25} - \frac{y^2}{9} = 1$ and $\frac{(x+1)^2}{25} - \frac{(y-2)^2}{9} = 1$.

Compare the two graphs and find the similarities and differences between the two hyperbolas. Find the foci and asymptotes for both hyperbolas.

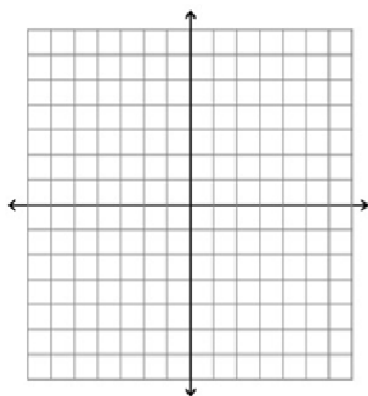


Problem 167. Use completing the square to identify type of each conic section. Graph each conic section. List the relevant items for each from this list: center, radius, vertex (plural?), focus (plural?), directrix and asymptotes.

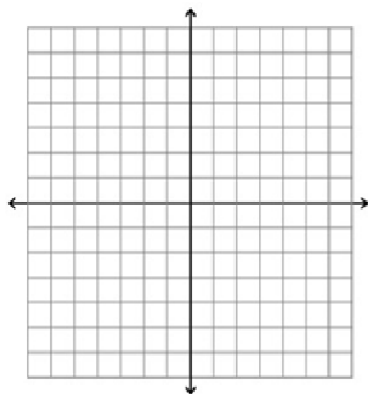
1. $x^2 - 4y^2 - 6x - 32y - 59 = 0$



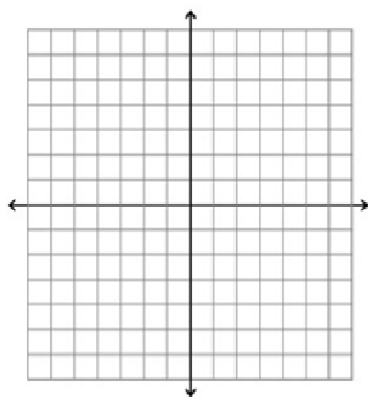
2. $9x^2 + 4y^2 + 18x - 16y - 11 = 0$



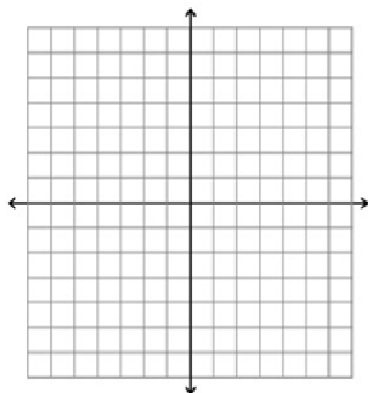
3. $x^2 + y^2 + 10 = 6y - 4 + 4x$



4. $x^2 + y^2 + 6x - 14y = 6$



5. $3x^2 - 2x + y = 5$



Chapter 4

Exponential Functions

Note: Just a reminder that:

1. An integer is a number in the set $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$.
2. The rationals are all numbers that can be written as $\frac{p}{q}$ where p and q are integers with $q \neq 0$.

Definition 31. Let b be any positive number aside from 1 and let n be any positive integer. The expression b^n is an **exponential expression** and means $b \times b \times \dots \times b$ where b is multiplied by itself n times. We now define a function $f(x) = b^x$ and call it an **exponential function** with **base** b and domain the positive integers.

Theorem 6. *Properties of Exponentials*

1. $b^x b^y = b^{x+y}$

2. $\frac{b^x}{b^y} = b^{(x-y)}$

3. $(b^x)^y = b^{(xy)}$

Problem 168. *Given any exponential function $f(x) = b^x$ with domain the positive integers, use Theorem 6 to extend $f(x)$ to a function $g(x) = b^x$ with domain all integers. This just means the domain of $g(x)$ is all the integers and if x is a positive integer then $g(x) = f(x)$.*

Problem 169. *Given any exponential function $g(x) = b^x$ with domain all integers, use Theorem 6 to extend $g(x)$ to a function $h(x) = b^x$ with domain all rational numbers. This just means the domain of $h(x)$ is all rational numbers and if x is any integer then $h(x) = g(x)$ and that the properties of exponentials developed with the integers are still true when extended to rational numbers.*

Note: Given any positive b aside from 1, we have now defined an exponential function $h(x) = b^x$ with domain all rational numbers. It turns out that these functions can be extended to domain of all real numbers. From this point on whenever we are given a positive number b aside from 1, then $f(x) = b^x$ is an exponential function with domain all real numbers.

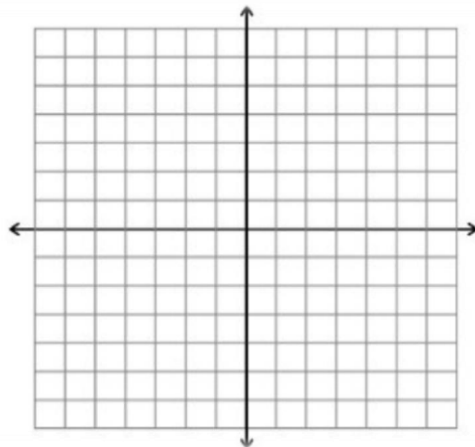
Problem 170. *Convince yourself that $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$ exists by substituting $n = 10, n = 100, n = 1000$ and $n = 10000$. Approximate this value to two decimal places.*

Definition 32. *We define the real number $e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$. The number e is known to be irrational and to two decimal places we write $e \approx 2.72$.*

Problem 171. *Create a table and graph each of the following exponential functions:*

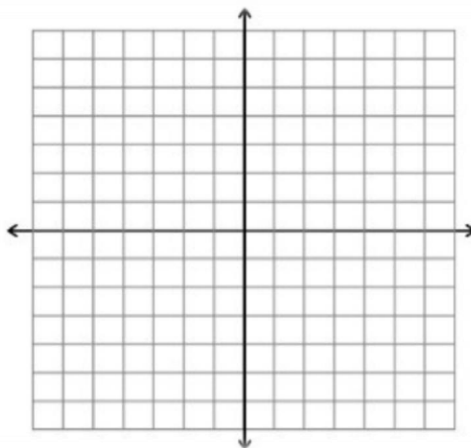
1. $y = f(x) = 2^x$

x	y



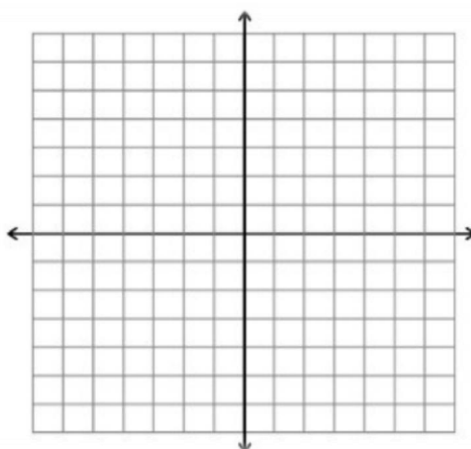
2. $y = g(x) = 3^x$

x	y



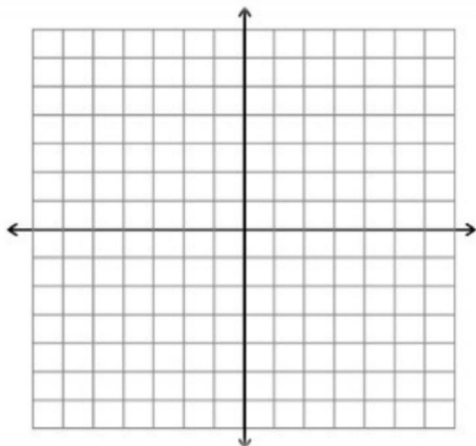
3. $y = h(x) = \left(\frac{1}{2}\right)^x$

x	y



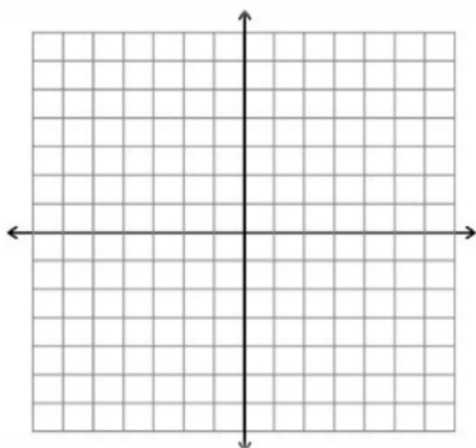
4. $y = k(x) = \left(\frac{1}{3}\right)^x$

x	y



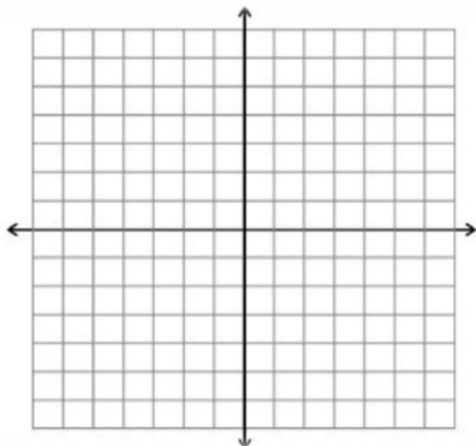
5. $y = w(x) = e^x$

x	y



6. $y = s(x) = e^{x-2}$

x	y



Problem 172. Consider the graphs from problem 171. For any positive base b , we know the domain is all real numbers. Can you make a claim about the range of $f(x) = b^x$? Describe differences between graphs depending on whether $b < 1$ or $b > 1$.

Problem 173. *Solve the following exponential equations:*

1. $3^x = 81$

2. $5^x = \left(\frac{1}{625}\right)$

3. $2^x - \frac{1}{8} = 0$

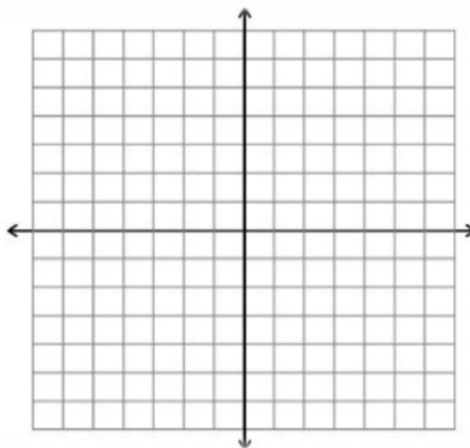
4. *Explain why $4^x + 5 = 0$ has no real number solution.*

Chapter 5

Logarithmic Functions

Problem 174. Consider the exponential function $y = f(x) = 2^x$ which was the first table and graph of problem 171. Create a new table by interchanging x and y . Graph this function. What is the domain and range of this function?

x	y



Definition 33. If $f(x) = b^x, b \neq 1$ is any exponential function, then we write the inverse function $f^{-1}(x) = \log_b(x)$. Thus, $\log_b(A) = k$ if and only if $b^k = A$. The inverse functions of exponential functions are called **logarithmic functions**.

Notation: There are two important logarithmic functions. First, since our number system is base 10, the exponential function 10^x is an important exponential function. We write this inverse function as $\log(x)$, not writing the 10 as a subscript to the log. Second, we have already developed the number e with the corresponding exponential function e^x . This exponential has many very surprising natural occurrences. Thus for $f(x) = e^x$, the inverse function is called the natural logarithm and is notated by $f^{-1}(x) = \ln(x)$.

Obviously there are many different exponential functions and hence many different logarithmic functions. When we say *THE* exponential function we mean $f(x) = e^x$, and when we say *THE* logarithmic function we mean $g(x) = \ln(x)$.

The following theorem gives justification for why we refer to the logarithmic function and the exponential function.

Theorem 7. Change of Base Theorem. For any positive integer b :

$$\log_b(x) = \frac{\ln(x)}{\ln(b)}$$

Theorem 8. *Properties of Logarithms.*

1. $\ln(xy) = \ln(x) + \ln(y)$

2. $\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$

3. $\ln(x^y) = y\ln(x)$

Problem 175. *Use the properties of logarithms to solve the following equations. Use your calculator to approximate answers to four decimal places.*

1. $2^x = 100$

2. $5^{(x-2)} = 500$

3. $e^{.05t} = 2500$

Problem 176. *Use properties of logarithms to rewrite each expression as a single logarithm.*

1. $\log_4(A) + \log_4(B)$

2. $\ln(100) - \ln(4)$

Problem 177. *Solve the following logarithmic equations for exact values.*

1. $\log_3(2x) = 4$

2. $\ln(1 - x) = 5$

3. $3\ln(3x - 5) - 1 = 0$

4. $\ln(x) + \ln(x - 1) = 0$

Chapter 6

Polynomial Functions

Definition 34. Given any non-negative integer n and any real numbers $a_n, a_{n-1}, a_{n-2}, \dots, a_1, a_0$ we define a **polynomial** with **degree** n to be:

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0$$

Definition 35. If $p(x) = k$ for some real number k , then we say $p(x)$ is a polynomial of degree 0, or that $p(x)$ is a **constant polynomial**. Polynomials of degree 1 are called **linear polynomials**. Degree 2 polynomials are called **quadratics**, degree 3 polynomials are called **cubics**, degree 4 polynomials are called **quartics** and degree 5 polynomials are called **quintics**.

Definition 36. Given a polynomial $p(x)$ and any number r such that $p(r) = 0$ we say that r is a **zero** of the polynomial $p(x)$. A zero of a polynomial is often referred to as a **root** of the polynomial. For any zero r of a polynomial, since $p(r) = 0$, the graph of $p(x)$ must go through $(r, 0)$, so r is also called an x – intercept of the polynomial.

Definition 37. If $x = r$ is a zero of a polynomial $p(x)$, then $x - r$ is a **linear factor** of the polynomial $p(x)$. Any polynomial of degree n can have at most n distinct zeros, thus at most n distinct linear factors.

Problem 178. *Earlier in this course we developed the quadratic formula to solve quadratic equations. Solve the following quadratics.*

1. $x^2 - x - 1 = 0$

2. $2x^2 - 3x - 5 = 0$

3. $x^2 + x + 1 = 0$

What do you notice about this solution?

6.1 Complex Numbers

Definition 38. *The solutions to the third part of problem 178 contains the square root of a negative number. There is no real number whose square is negative, thus the solutions cannot be real numbers. This difficulty is solved by making the following definition: Define $\sqrt{-1} = i$. Then $i^2 = -1$.*

Problem 179. *Show that any number written as $a + \sqrt{-b}$ where a and b are real numbers can be written as $a + \sqrt{b}i$. Write the solutions from the third part of problem 178 in the form $a + bi$ where a and b are real numbers.*

Definition 39. The set of all numbers that can be written in the form $a + bi$ with a and b real numbers is called the set of **complex numbers**. If $z = a + bi$ is any complex number, then we call a the real part of z and write $\text{Re}(z) = a$ and b the imaginary part of z and we write $\text{Im}(z) = b$.

Definition 40. For any complex number $z = a + bi$, the **conjugate** of z is the complex number $a - bi$. Conjugate pairs are very important. Note that the solutions from problem 178 are conjugate pairs! Also note that any real number r can be written as $r + 0i$, thus r is a complex number.

Notation: Addition, subtraction, multiplication and division can all make sense with complex numbers. Addition and subtraction are performed component-wise on the real numbers that comprise the complex numbers. Multiplication can be performed using familiar FOIL method used to multiply binary expressions in algebra. Conjugate pairs with simplification can be used to divide complex numbers.

Problem 180. Let z_1 and z_2 be any two complex numbers.

1. Show that $z_1 + z_2$ is a complex number. Show that $z_1 - z_2$ is a complex number.
2. Show that $z_1 z_2$ is a complex number by multiplying by FOIL method.
3. Show that $\frac{z_1}{z_2}$ is a complex number by multiplying both numerator and denominator by the conjugate of z_2 and simplifying.

Problem 181. *Show that if z_1 and z_2 forms a conjugate pair then $z_1 z_2$ is a real number.*

Problem 182. *Perform the following operations with complex numbers.*

1. $(1 - i) + (3 + 5i)$

2. $(1 - i) - (8 - 2i)$

3. $(2 - 2i)(3 - i)$

4. $(5 - i)(7 - 2i)$

5. $\frac{3i}{1+i}$

6. $\frac{1+i}{2-3i}$

Note: Any polynomial with positive degree n has exactly n zeros, though not necessarily distinct. All zeros of any polynomial are complex numbers. Remember, any real number r is a complex number with $\text{Im}(r) = 0$. If $a + bi$ is a zero of the polynomial $p(x)$, then $a - bi$ must also be a zero of the polynomial $p(x)$. In other words, complex numbers show up as zeros of polynomials in conjugate pairs.

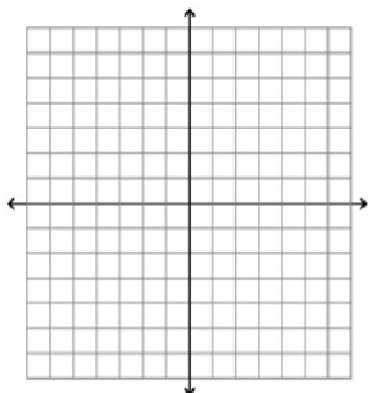
Definition 41. When considering the graph of a function, the **end behavior** of the function is the behavior of the graph as $x \rightarrow \pm\infty$. Thus, the end behavior is the value that the function approaches arbitrarily closely as x increases or decreases without bound. Note that if no such finite value exists, then the end behavior of the function is infinite.

Problem 183. Let $p(x)$ be any polynomial with even degree n and $n \geq 2$. Describe all possible end behaviors for the graph of $p(x)$ giving plausible explanations for any claims made.

Problem 184. Let $p(x)$ be any polynomial with odd degree n and $n \geq 1$. Describe all possible end behaviors for the graph of $p(x)$ giving plausible explanations for any claims made.

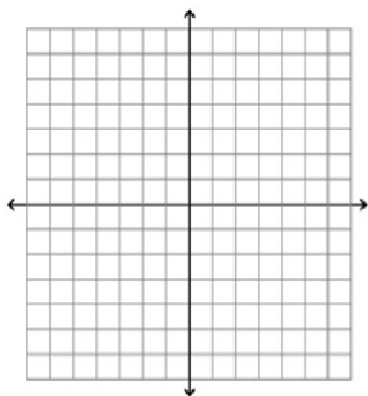
Problem 185. Draw any feasible graph for polynomial $p(x)$ that satisfies the following requirements.

1. The degree of $p(x)$ is 5.
2. $p(x)$ has exactly three real zeros.



Problem 186. Draw any feasible graph for polynomial $p(x)$ that satisfies the following requirements.

1. The degree of $p(x)$ is 7.
2. $p(x)$ has exactly two non-real complex zeros.
3. The leading coefficient is $a_7 < 0$.



The following theorem is stated, but not proved.

Theorem 9. Suppose $p(x)$ is a polynomial with:

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0 = 0$$

Then all rational zeros of $p(x)$ are of the form $\frac{p}{q}$ satisfying:

1. p is an integer factor of the constant coefficient a_0 .
2. q is an integer factor of leading coefficient a_n .

Problem 187. Find all rational zeros of the following polynomials, then factor each into linear factors. Note: Some factors could be of form $(x - z)$ where $z = a + bi$ and $b \neq 0$. Linear factors of form $z = a + bi$ and $b \neq 0$ are NOT polynomials because all coefficients of polynomials are real numbers. Hint: You may need to perform long division.

1. $p(x) = x^4 - 2x^3 - 5x^2 + 4x + 6$

2. $p(x) = x^4 - 2x^3 - x^2 - 4x - 6$

Chapter 7

Rational Functions

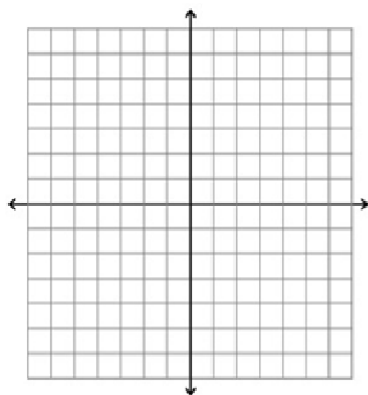
Note: In this chapter we will regularly use the fact that $\frac{1}{x} \rightarrow 0$ as $x \rightarrow \infty$.

Definition 42. Let $p(x)$ and $q(x)$ be any two polynomial functions.

$$h(x) = \frac{p(x)}{q(x)}, \quad q(x) \neq 0$$

Then $h(x)$ is called a **rational function**. Whenever $q(c) = 0$ then $h(x)$ is not defined at the point $x = c$. Given $h(x) = \frac{p(x)}{q(x)}$ with $p(x)$ and $q(x)$ having no common polynomial factor, then the graph of $y = h(x)$ has **vertical asymptotes** $x = c$ for all c such that $q(c) = 0$.

Problem 188. Explain why the graph of the rational function $h(x) = \frac{x^2-9}{x-3}$ does not have a vertical asymptote at $x = 3$. Graph $y = h(x)$.

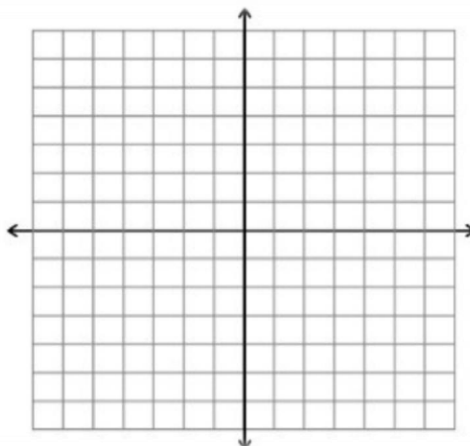


Problem 189. Let $h(x) = \frac{f(x)}{g(x)}$ be any rational function such that $f(x)$ and $g(x)$ have no common polynomial factors. If the degree of $f(x)$ is strictly less than the degree of $g(x)$ then show that the graph of $h(x)$ has the x -axis as a horizontal asymptote and that there is no other horizontal asymptote.

Problem 190. Let $h(x) = \frac{f(x)}{g(x)}$ be any rational function such that $f(x)$ and $g(x)$ have no common polynomial factors. If the degree of $f(x)$ is equal to the degree of $g(x)$ then show that the graph of $h(x)$ has a horizontal asymptote at $y = \frac{a_n}{b_n}$ where a_n is the leading coefficient of $f(x)$ and b_n is the leading coefficient of $g(x)$. Also show that this is the only horizontal asymptote.

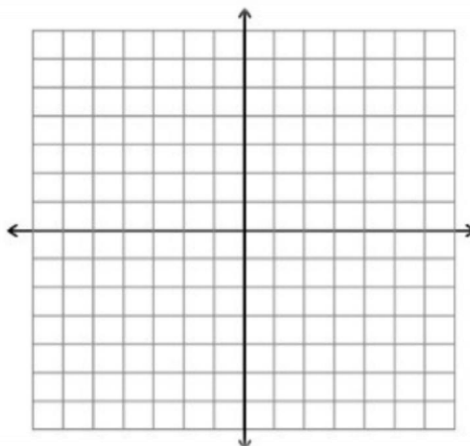
Problem 191. Find all vertical and horizontal asymptotes for $h(x) = \frac{x^2 - 2x - 3}{x^2 - 1}$. Create a table and graph for $h(x)$.

x	y



Problem 192. Find all vertical and horizontal asymptotes for $h(x) = \frac{x^2-1}{x^3-1}$. Create a table and graph for $h(x)$.

x	y



Problem 193. Show that the rational function $h(x) = \frac{x^2+2x+1}{x-1}$ has a non-horizontal linear asymptote. Find the equation of this linear asymptote.

Chapter 8

Linear Functions

Definition 43. Given the polynomial $p(x) = a_1x + a_0$, then $p(x)$ is called a **linear function**. Linear functions are often written as:

$$y = mx + b$$

This should be familiar from High School Algebra.

m is called the **slope** of the line and is the rate of change of y over the rate of change of x , or more familiarly, given two points (x_1, y_1) and (x_2, y_2) , then:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

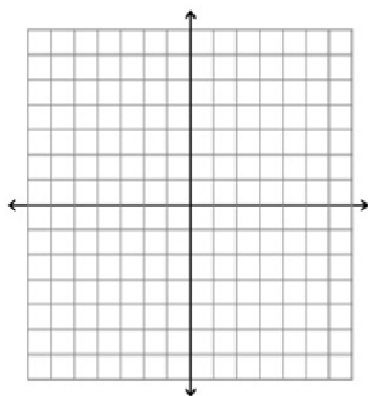
Problem 194. Establish that the following lines are perpendicular.

$$y_1 = mx + b_1$$

$$y_2 = -\frac{1}{m}x + b_2$$

Problem 195. *Do the following.*

1. *Draw any line through the origin. Call the slope m so that the equation of your line is $y = mx$.*
2. *Plot any non-origin point. Call this point (h, k) .*
3. *Use vertical and horizontal translations to find an equation for the line with slope m through the point (h, k) .*



Notation: Problem 195 developed what is known as the point slope form for the equation of a line. This is usually written as:

$$(y - y_1) = m(x - x_1)$$

You will find this equation very useful as you start Calculus.

Problem 196. *Find equation of line through the points $(2, -1)$ and $(5, 2)$. Also find equation of a line through $(5, 2)$ perpendicular to the line you just found.*

8.1 Average Rate of Change

Problem 197. *A particle moves away from point A. The distance of the particle from point A after t seconds is given by:*

$$s(t) = 2t^2 + t + 5$$

1. *How far is the particle from the point A after 1 second?*
2. *How far is the particle from point A after 3 seconds?*
3. *What is the average velocity of the particle over the time interval from 1 second to 3 seconds?*
4. *Confirm that the average velocity is the slope of the line through the points $(1, s(1))$ and $(3, s(3))$.*

Problem 198. *This problem will generalize ideas from problem 197. A particle moves from point A. The distance of the particle from point A after t seconds is given by $s(t)$.*

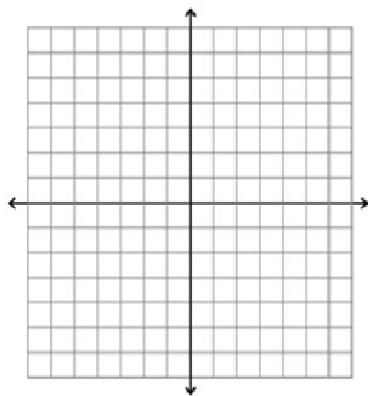
1. *How far is the particle from point A after a seconds?*
2. *How far is the particle from point A after b seconds?*
3. *What is the average velocity of the particle?*

Definition 44. Problems 197 and 198 developed the idea of average velocity which is just average rate of change. Given any function $f(x)$, the **average rate of change** of $f(x)$ over the interval $[a, b]$ is given by:

$$\text{Average rate of change} = \frac{f(b) - f(a)}{b - a}$$

This is just the slope of the line through the points $(a, f(a))$ and $(b, f(b))$.

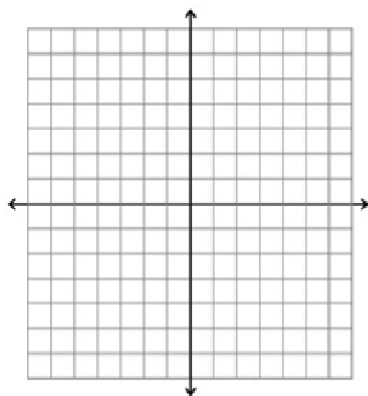
Problem 199. Graph a circle of radius 2 units centered at the origin. There are two lines tangent to the circle with slope $m = -1$. Draw these two lines and their corresponding normal lines. Find equations for all four of these lines.



Problem 200. Draw the circle represented by:

$$(x - 2)^2 + (y - 3)^2 = 4$$

Repeat problem 199 with this circle.



Essay 5. Write an essay describing your experience in this course. Will anything about this course have a lasting impact on you? This essay is due at the final. By turning this essay in on time you will receive full points as a grade for the essay. The essay will not be read until after final grades are posted.

Notes

¹(Definition 1): I personally like to give a very brief historical talk about Rene Descartes's innovative ideas. Linking equations to geometric objects was not an obvious thing to do!

²(Theorem 1) This theorem is done with an entire class discussion.

³(Theorem 3) It is best not to let this theorem sit too long. If students appear to be making no progress, a good hint is to use parallel lines cut by a transversal.

⁴(Theorem 4) It is best not to let this theorem sit too long. A class discussion and perhaps even a mini lecture can be very useful here. Emphasis needs to be placed on the fact that, mathematically, this theorem is very accessible to the students. The only difficulty is the ability to leverage their knowledge to prove a very intriguing concept that the students will shortly see has many applications.

⁵(Problem 29) This problem reveals a usual algebraic weakness. Students often have a very difficult time coordinating units. Good idea to have a mini-lecture on proportionality and dimension analysis. Appropriate quizzes may help students gain needed insights.

⁶(Note after problem 32) I have found this to be a critical junction point in the course. This is a good point for a good comprehensive review of all topics covered, specifically mentioning which topics will be covered on exams. Students should know to read the following definitions carefully, as they are a direct link to trigonometric functions.

⁷(Definition 12) This is an appropriate time to emphasize the importance of the concept of function in mathematics. The class should discuss the various ways of defining the function concept.

⁸(Problem 39) This is a good place to have a class discussion regarding the idea of notating an infinite solution set. Two main points need to be understood:

1. The students must be held accountable to the convention they agree upon to notate an infinite solution set.
2. The students need to understand that if they enter another community of people doing mathematics, then they may need to adapt another convention to notate an infinite solution set. There are many ways to communicate mathematical ideas!

⁹(Definition 18) Time well spent discussing this definition in detail.

¹⁰(Problem 75) After this problem is done, a mini lecture discussing how a pentagon construction can yield exact value for $\sin(36^\circ)$.

¹¹(Problem 86) Solving this problem by squaring both sides results in unwanted solutions. This is a good time for a mini-lecture and class discussion concerning extraneous solutions and why they occur.

¹²(Theorem 5) This is a good time to review importance of completing the square process for solving quadratic equations. Most students have seen this, but missed the significance. Students should be reminded that this concept will show up again when studying conic sections.